

ATOMIC STRUCTURE

Passage-1

de Broglie proposed dual nature for electron by putting his famous equation $\lambda = \frac{h}{mu}$. Later on

Heisenburg proposed uncertainty principle as $\Delta p \cdot \Delta x \geq \frac{n}{2} \left(n = \frac{h}{2\pi} \right)$. On the contrary particle nature of electron was established on the basis of photoelectric effect. When a photon strikes the metal surface, it gives up its energy to the electron. Part of this energy (say W) is used by the electrons to escape from the metal and the remaining imparts the kinetic energy ($\frac{1}{2} mu^2$) to the photoelectron. The potential applied on the surface to reduce the velocity of photoelectron to zero is known as stopping potential.

- The wavelength of helium atom whose speed is equal to its rms speed at $27^\circ C$:
 (A) $7.29 \times 10^{-11} m$ (B) $4.28 \times 10^{-10} m$ (C) $5.31 \times 10^{-11} m$ (D) $6.28 \times 10^{-11} m$
- With what potential should a beam of electron be accelerated so that its wavelength becomes equal to $1.54 \text{ \AA}$:
 (A) 63.3V (B) 6.33V (C) 633V (D) None of these
- The wavelength of a golf ball weighing 200g and moving at a speed of 5m/hr is of the order
 (A) $10^{-10} m$ (B) $10^{-20} m$ (C) $10^{-30} m$ (D) $10^{-40} m$

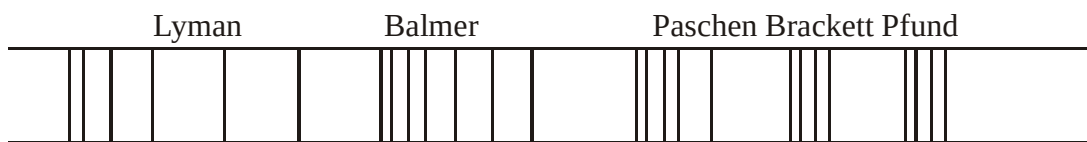
Passage-2

The only electron in the hydrogen atom resides under ordinary conditions on the first orbit. When energy is supplied, the electron moves to higher energy shells depending on the amount of energy absorbed. When this electron returns to any of the lower energy shells, it emits energy. Lyman series is formed when the electron returns to the lowest energy state while Balmer series is formed when the electron returns to second energy shell. Similarly, Paschen, Brackett and Pfund series are formed when electron returns to the third, fourth and fifth energy shells from higher energy shells respectively. Thus, the different spectral lines in the spectra of atoms correspond to different transitions of electrons from higher energy levels to lower energy levels.

- If the shortest wavelength of H atom in Lyman series is x , then longest wavelength in Balmer series of He^+ is :
 (A) $\frac{9x}{5}$ (B) $\frac{36x}{5}$ (C) $\frac{x}{4}$ (D) $\frac{5x}{9}$
- The ratio of the number of spectral lines obtained when an electron jumps from 7th level to ground state to 6th level to 3rd level.
 (A) 7 (B) 3.5 (C) 2.5 (D) 10
- In hydrogen atom which of the following transitions should be associated with highest absorption of energy?
 (A) $n = 1$ to $n = 4$ (B) $n = 2$ to $n = 3$ (C) $n = 4$ to $n = 1$ (D) $n = 3$ to $n = 2$

Passage-3

If Hydrogen gas enclosed in a sealed tube is heated to high-temperature, it emits radiation, if this radiations is passed through a prism, components of different wavelengths are deviated by different amounts and thus we get the Hydrogen-spectrum. The most striking feature in this spectrum is that is only some sharply defined, discrete wavelengths exist in the emitted radiations. A hydrogen sample also emits radiations with wavelengths less than those in visible range and also with wavelength larger than those in visible range. We get that x lines may be grouped in seperate series i.e lyman series, Balmer, series, paschen, Brackett, pfund.



(a) Balmer's Law. The actual development of the spectroscopy started in 1885 when J.J. Balmer found that the wave-length of the hydrogen lines could be represented by the formula:

$$\lambda = B \left(\frac{n^2}{2^2 - n^2} \right)$$

where B is a constant, n is an integer with variable values 3,4,5,..... etc.

7. In which of the following transistion energy change is minimum
 (A) $1 \rightarrow 2$ (B) $2 \rightarrow 3$ (C) $4 \rightarrow 5$ (D) $3 \rightarrow 4$
8. What is the wavelength of radiation emitted when the e⁻s in a hydrogen atom jumps from $n = \infty$ to pfund series
 (A) 6463 nm (B) 2283 nm (C) 3650 nm (D) 8265 nm
9. Difference of wavelength of extreme lines for Lyman Series is
 (A) $\frac{1}{3R}$ (B) $\frac{5}{36R}$ (C) $\frac{3}{4R}$ (D) $\frac{1}{R}$

Passage-4

The French physicist Louis de Broglie in 1924 postulated that matter, like radiation, should exhibit a dual behaviour. He proposed the following relationship between the wavelength λ of a material particle, its linear momentum p and planck constant h

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

The de-Broglie relation implies that th wavelength of a particle should decreases as its velocity increase. It also implies that for a given velocity heavier particles should have shorter wavelength than lighter particles. The waves associated with particles in motion are called matter waves or de Broglie waves. These waves differ from the electromagnetic waves as they

- (i) have lower velocities
- (ii) have no electrical and magnetic fields and
- (iii) are not emitted by the particle under consideration.

The experimental confirmation of the de Broglie relation was obtained when Devisson and Gerner, in 1927, observed that a beam of electron is diffracted by a nickel crystal. As diffraction is a characteristic property of waves, hence the beam of electron behaves as a wave, as proposed by de Broglie.

Warner Heisenberg considered the limits of how precisely we can measure properties of an electron or other microscopic particle like electron. He determined that there is a fundamental limit of how closely we can measure both positon and momentum. The more accurately we measure the momentum of a particle, the les accurately we can determine its position. The

converse is also true. This is summed up in what we now call the 'Heisenberg uncertainty principle'. It is impossible to determine simultaneously and precisely both the momentum and position of a particle. The product of uncertainty in the position, Δx and the uncertainty in the momentum

$\Delta(mv)$ must be greater than or equal to $\frac{h}{4\pi}$ i.e.

$$\Delta x \Delta(mv) \geq \frac{h}{4\pi}$$

10. If the uncertainty in velocity and position is same, then the uncertainty in momentum will be:

(A) $\sqrt{\frac{hm}{4\pi}}$ (B) $m\sqrt{\frac{h}{4\pi}}$ (C) $\sqrt{\frac{h}{4\pi m}}$ (D) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

11. Two particles A and B are in motion. If the wavelength associated with the particle A is $5.0 \times 10^{-8} \text{ m}$, the wavelength of particles B having momentum half of A is

(A) $2.5 \times 10^{-8} \text{ m}$ (B) $1.25 \times 10^{-8} \text{ m}$ (C) $1.0 \times 10^{-7} \text{ m}$ (D) $1.0 \times 10^{-8} \text{ m}$

12. The uncertainty in the position of an electron (mass = $9.1 \times 10^{-28} \text{ g}$) moving with a velocity of $3.0 \times 10^4 \text{ cm s}^{-1}$ accurate upto 0.011% will be

(A) 1.92 cm (B) 0.768 cm (C) 0.175 cm (D) 3.84 cm

Passage-5

The behaviour of an electron in an atom is described mathematically by a wave function, or orbital. It turns out that each wave function contains three variables, called quantum numbers, which are represented as n, l and m_l . These quantum numbers describe the energy level of an orbital and define the shape and orientation of the region in space where the electron will be found.

13. Which quantum number determines orientation of the electron ?

(A) Principal (B) Secondary (C) Magnetic (D) Spin

14. Radial nodes are maximum in

(A) 4s (B) 4p (C) 3d (D) 5f

15. Consider following statements.

A. Splitting of spectral line occurs when placed in a magnetic field or in an electric field

B. In case of 1s-orbital, the density of the charge cloud is the greatest at the nucleus and falls off with the distance. The density (at a particular distance) is uniform

C. Electron-density is concentrated along a particular direction in case of 2p-orbital.

D. A p-orbital can take maximum of six electrons.

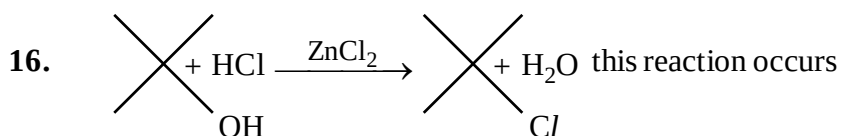
Select the correct option

(A) A, B, D (B) A, B, C (C) B, C, D (D) A, C, D

Passage-6

Alcohol and considered as a best starting material far preparing alkyl halides. Most common reagent used for this are HX, PCl_5 , PCl_3 , SOCl_2 . Among the halogen acid the order of reactivity is $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ & for Alcohol $3^\circ > 2^\circ > 1^\circ$ $\text{CH}_3\text{-OH}$.

In the case of 1° Alcohol when we use HX rearrangement occurs and mixture of product form but if we use PCl_3 or SOCl_2 it occurs v/a S_N^2 .



(A) within a minutes

(B) when temperature is $0 - 100^\circ\text{C}$

(C) after 5 minutes

(D) Reaction does not take place

17. Which of the following is not true

(A) SOBr_2 is less stable and SOI_2 does not exist

(B) In preparing alkyl halide from Alcohol the best method is application of SOCl_2

(C) 3° Alcohol react with HX via S_N^1 pathway.

(D) optically active alcohol (1°) react with PCl_3 give the racemic mixture

18. $\text{R-OH} + \text{A} \rightarrow \text{R-Cl}$

Which of the following should be taken as A to give the highest yield & no separation of product will required ?

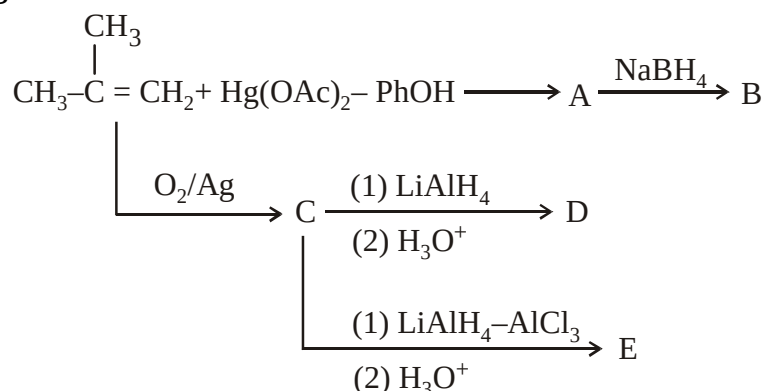
(A) PCl_5

(B) PCl_3

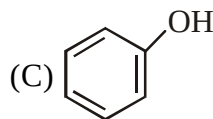
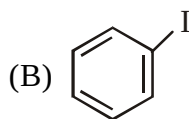
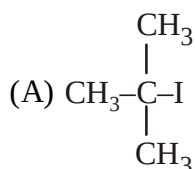
(C) SOCl_2

(D) HCl/ZnCl_2

Passage-7

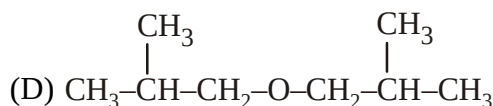
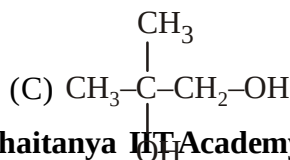
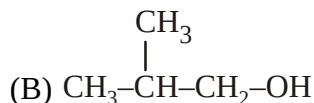
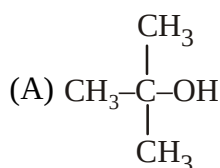


19. The compound B $\xrightarrow{\text{HI}}$

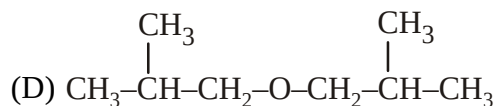
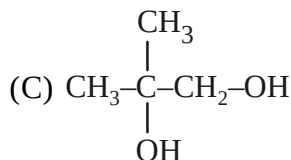
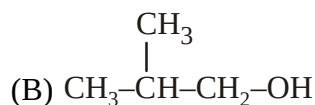
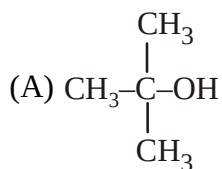


(D) Both (A) and (C)

20. The compound D is:

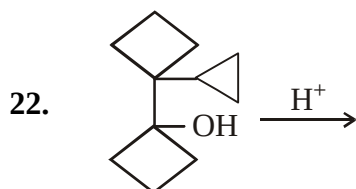


21. The compound E is:

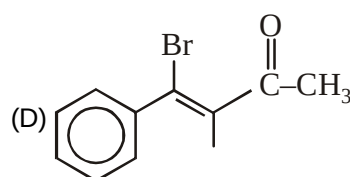
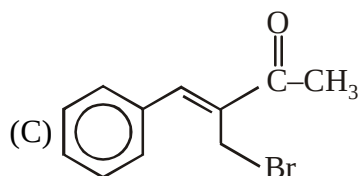
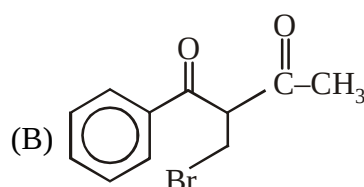
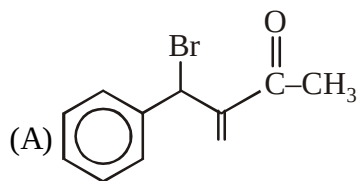
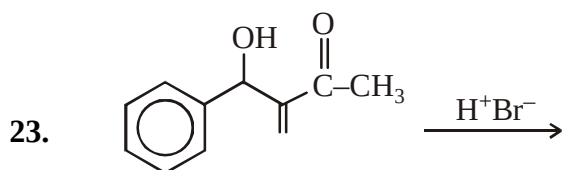
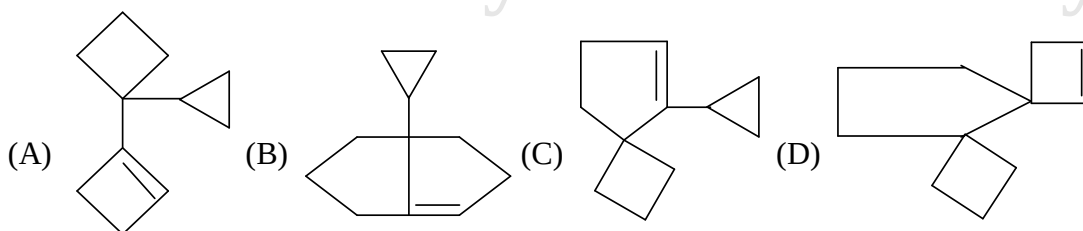


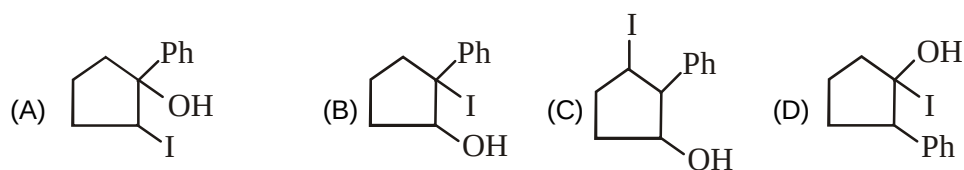
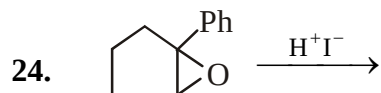
Passage-8

Protonation of organic compounds gives a lots of product having great synthetic application. Alcohols, alkenes and many other groups react with H^+ ion to give carbocation which may undergo rearrangement to form stable carbocation. In general the stability of carbocation is $3^\circ > 2^\circ > 1^\circ$



Rearranged alkene product after rearrangement at low temperature will be mainly



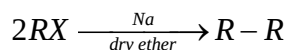


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ALKANES

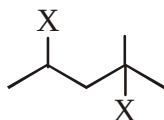
Passage-9

Alkanes can be prepared by Wurtz reaction, in which, alkyl halides are coupled each other with the help of metallic sodium in dry ether solvent, to give alkanes.



Due to elimination and rearrangement reactions, different byproducts are formed. Unsymmetrical alkanes can be prepared by this method, but practically it is very difficult to separate the individual alkane from the mixture of alkanes.

25. Which one of the following is correct for the following dihalide:

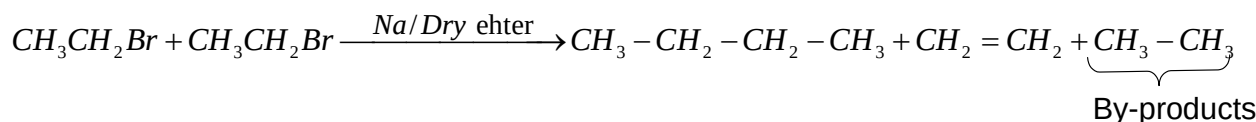


- (A) intramolecular coupling alone take place
 (B) extramolecular coupling alone take place
 (C) intramolecular coupling dominates extramolecular coupling
 (D) extramolecular coupling dominates intramolecular coupling
26. Identify the incorrect statement. In the stoichiometric Wurtz reaction on ethyl chloride.
 (A) disproportionation of alkyl carbanion and alkyl halide take place.
 (B) some sodium metal remains in the reaction mixture
 (C) the byproducts dominate the chief product
 (D) inter molecular hydrogenation take place
27. The intermediate in Wurtz reaction is:
 (A) Carbanion (B) Carbocation
 (C) anion radical (D) organo metallic compound

Passage-10

Wurtz reaction involves the condensation of two molecules of alkyl halides in the presence of sodium and dry ether $R-X + 2Na + X-R \xrightarrow{\text{Dry Ether}} R-R + 2NaX$

In this reaction small amount of alkene is also formed as by-product.



Tertiary alkyl halides do not give Wurtz reaction. Frankland reaction is similar but has certain advantages over Wurtz reaction. It is useful in the synthesis of symmetrical alkanes.

Frankland reaction is shown by primary, secondary as well as tertiary alkyl halide.

Answer the following questions

28. Which of the following alkanes is not obtained from Wurtz reaction?
 (A) Methane (B) Ethane (C) Propane (D) Butane

29. A mixture of ethyl iodide and methyl iodide is subjected to the Wurtz reaction. The products formed are:
 (A) ethane (B) butane (C) propane (D) 2-methylpropane
30. The intermediate compound(s) formed in Frankland reaction is/are:
 (A) $RZnI_2$ (B) R_2Zn (C) $RZnI$ (D) R_2ZnI

Passage-11

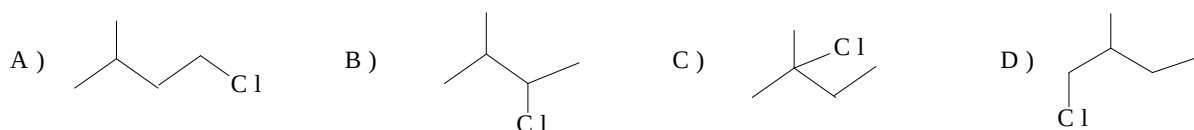
Chlorination on alkanes at below $300^\circ C$

Reactivity order $3^\circ - H > 2^\circ - H > 1^\circ - H$

4.5 : 3.25 : 1

mono chlorination on 2-methyl butane

31. In the above reaction major product is



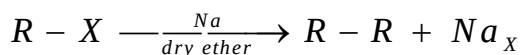
32. Number of isomers obtained in the above reaction

(A) 4 (B) 3 (C) 6 (D) 5

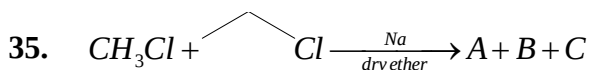
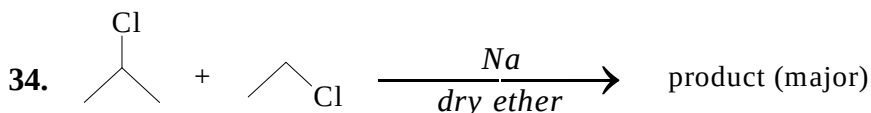
33. If chlorination takes place at '1' and '3' positions on 2-methyl butane simultaneously then how many optically active isomers are possible.

(A) 2 (B) 4 (C) 6 (D) 8

Passage-12



Mechanism is uncertain but explained by both ionic and free radical mechanism.



A, B, C are saturated hydrocarbons. Then A, B, C are

(A) Chain isomers (B) Positional isomers (C) Homologous (D) Diastereomers

36. Which of the following reactant is not suitable for Wurtz reaction

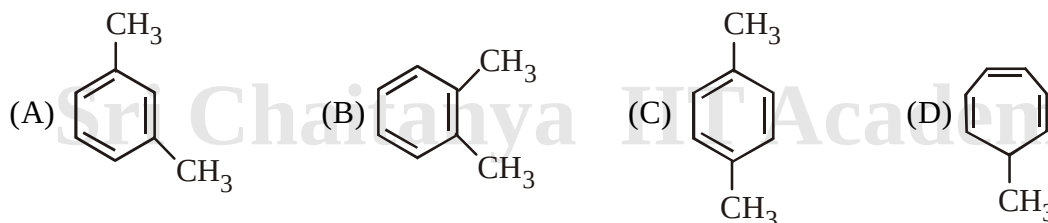


41. In case of Wurtz reaction which alkyl halide gives maximum yield of alkane?
 (A) Isopropyl bromide (B) Methyl bromide
 (C) Tert-butyl bromide (D) Ethyl bromide
42. To form the isopentane which of the following alkyl halides should be used in Corey house synthesis (RX) and (R'X) respectively as shown above :
 (A) Both $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{Br}$ (B) CH_3Br and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
 (C) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{Br}$ and CH_3Br (D) CH_3Br and $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{Br}$

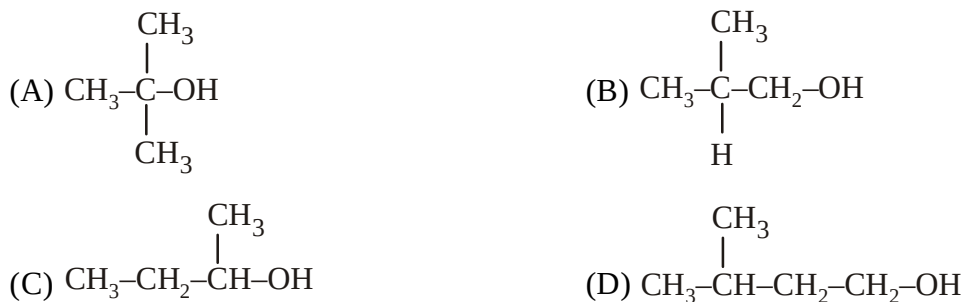
Passage-15

Two unknown compounds (A & B) have same molecular formula $\text{C}_4\text{H}_9\text{Br}$ which give n-butane on reaction with Zn-Cu/EtOH

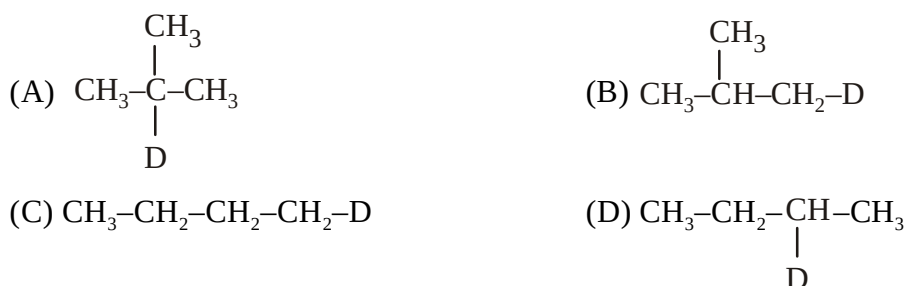
43. The compound A on wurtz reaction gives a linear hydrocarbon (X) which on further reaction with $\text{Cr}_2\text{O}_3/\text{Al}_2\text{O}_3$ gives an hydrocarbon that has octane number more than 100. The compound is:



44. The compound A $\xrightarrow{\text{LiAlH}_4}$ C $\xrightarrow{\text{AlCl}_3}$ D $\xrightarrow[\text{H}^+]{\text{KMnO}_4}$ E. The structure of E is:



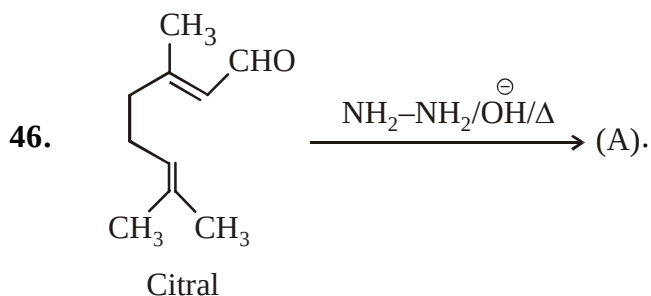
45. The compound B $\xrightarrow[\text{Et}_2\text{O}]{\text{Mg}}$ Y $\xrightarrow{\text{D}_2\text{O}}$ Z. The compound Z is



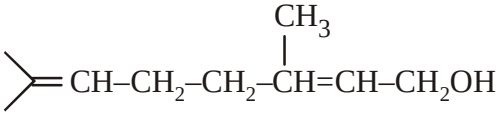
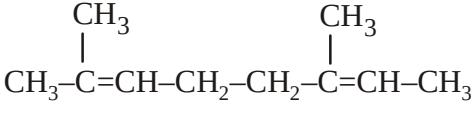
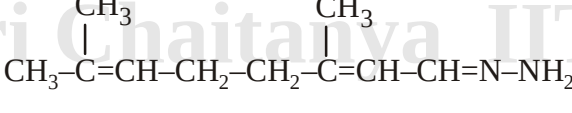
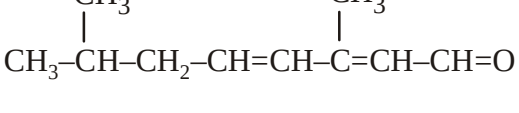
ALKENES

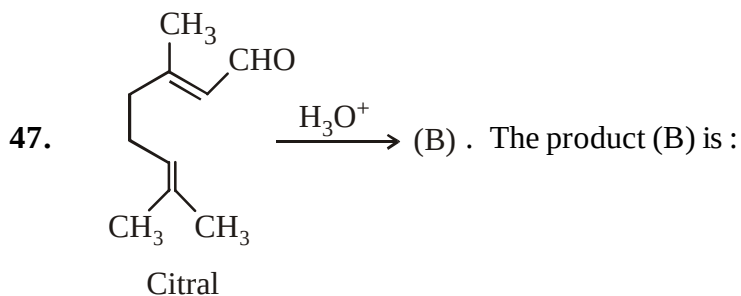
Passage-16

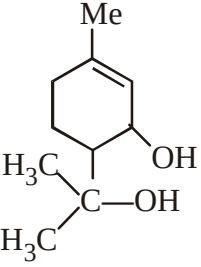
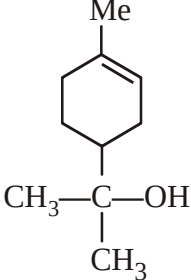
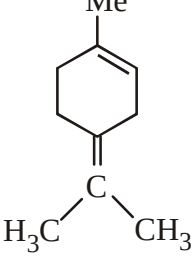
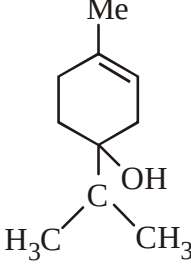
Citral is unsaturated aldehyde found in lemon oil. Fragrance of citral leaves and fruits is due to the presence of this compound.

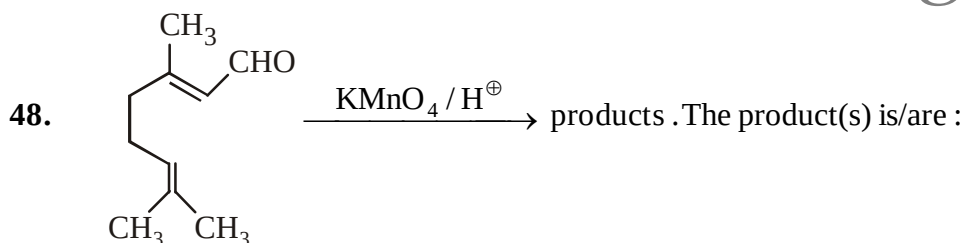


The product (A) is :

- (A) 
- (B) 
- (C) 
- (D) 



- (A) 
- (B) 
- (C) 
- (D) 



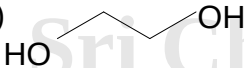
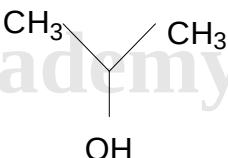
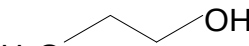
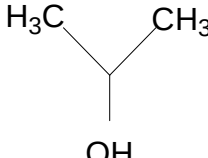
Citral

- (A) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})-\text{OH}$ (B) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ (C) CO_2 (D) All of these

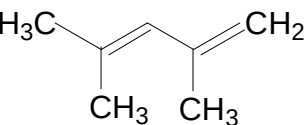
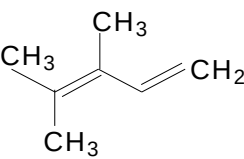
Passage-17

Alkenes can be oxidized in the presence of different reagents to give different products. Alkenes on oxidation in presence of silver at high temperature give cyclic ethers, which also can be synthesized by using per acids. Alkenes on reductive ozonolysis give carbonyl compounds in presence of Zinc and H_2O_2 . On reaction with acidic KMnO_4 , it also produces corresponding Carbonyl compound. Alkynes can be also oxidized in presence of such reagents but the products are different.

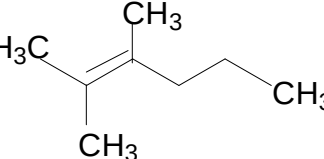
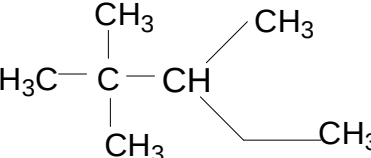
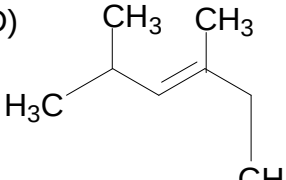
49. Ethylene on reaction with mCPBA (m-chloro per Benzoic acid) in CH_2Cl_2 forms a compound (X), which on reaction with CH_3MgI and subsequent Hydrolysis gives a compound (Y). (X) and (Y) are respectively

- (A)  and  (B)  and 
(C)  and  (D) $\text{H}_3\text{C}-\text{CHO}$ and 

50. An open chain Hydrocarbon (C_7H_{12}) on ozonolysis produces propanone ; methanal and 2-oxo-propanal. The Hydrocarbon could be

- (A)  (B) 
(C) Either of these (D) $\text{CH}_3-\text{CH}_2-\text{COOH}$

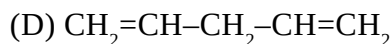
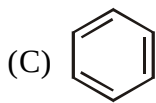
51. A Hydrocarbon (C_8H_{16}) on oxidation with a hot acidified solution of KMnO_4 forms 2-Butanone and Isobutyric acid as a product. The Hydrocarbon is

- (A) $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_2\text{CH}_3)_2$ (B) 
(C)  (D) 

Passage-18

An alkene (A) on oxidation with $K_2Cr_2O_7/H_2SO_4$ gives only two moles of gas (X), which turns lime water milky.

52. The 'A' can be:



53. 'B' that is next homologous of 'A'. The product/s on oxidation of 'B' are:

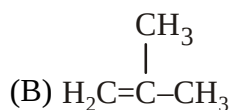
(A) Acetic acid

(B) Acetic acid and formic acid

(C) Acetone and acetic acid

(D) Acetic acid/ CO_2

54. The compound 'C' is the just higher homologous of 'B' but on oxidation does not evolve CO_2 . The 'C' is

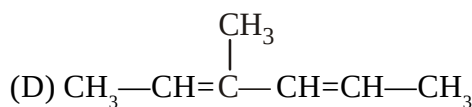
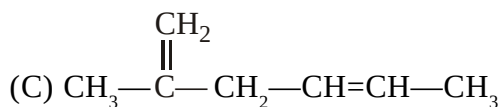
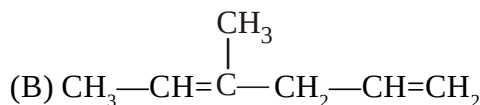
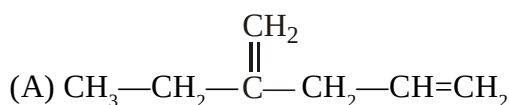


(D) None of these

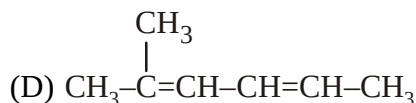
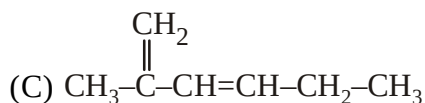
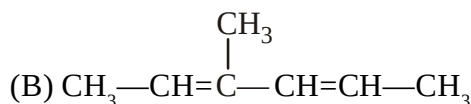
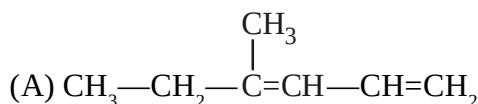
Passage-19

An unsaturated hydrocarbon A (C_7H_{12}) absorbs 2 molecule of hydrogen when hydrogenated. On oxidation it gives one molecule each of acetic acid, and acetoacetic acid and on reduction give 2-methylhexane :

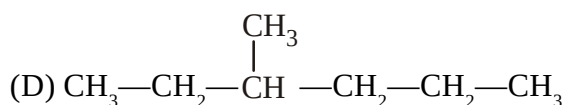
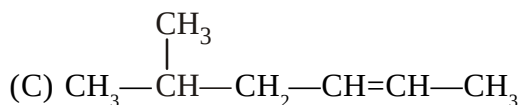
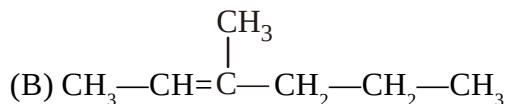
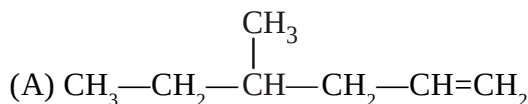
55. The structure of A:



56. The compound A on heating produces a more stable compound:

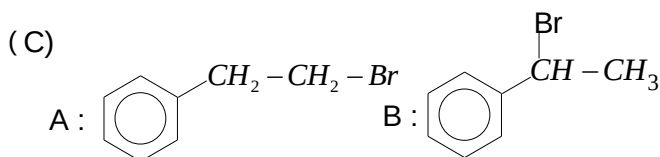
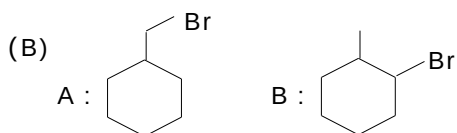
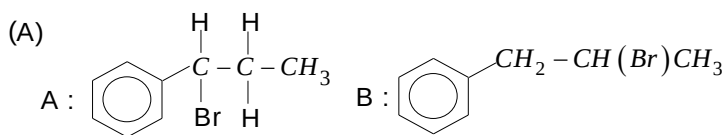


57. The A on Birch Reduction with $Na/Lq.NH_3$ gives:



Dehydrobromination of (A) & (B) gives same alkene (C). Alkene (C) can regenerate (A) & (B) by the addition of HBr in presence & absence of peroxide respectively. 1, 1 – diphenyl ethane is obtained on reaction of (C) with benzene in the presence of H^+ ions.

58. Compound 'A' & 'B' are



(D) None

59. Hydrolysis of (A) & (B) gives isomeric products (D) & (E). Which of the following statement for D & E is not true ?

- (A) Both D & E gives iodoform test
- (B) Both D & E gives same alkene upon dehydration
- (C) Both D & E reacts with NaOH
- (D) None of these

60. A when treated with NBS & then with NaNH_2 gives (X), when (X) is reacted with $\text{H}_2\text{O} / H^+$ it gives :-

- (A) Phenol
- (B) Acetophenone
- (C) Benzoic acid
- (D) None of these

CHEMICAL BONDING

Passage-21

In the ionic bond, the cation tend to polarise the electron cloud of the anion by pulling electron density towards itself. This causes development of covalent character in ionic bond because the electron density gets localised in between the nuclei. The tendency of cation to bring about the polarisation of anion is expressed as its polarising power. The ability of ion to undergo polarisation is called its polarisability. The polarising power of cation and polarisability of anion are decided on the basis of Fajan rules as given below :

- (i) Smaller the cation higher is its polarising power.
- (ii) Cations with pseudo noble gas configuration ($ns^2np^6nd^{10}$) have relatively high polarising power than those with noble gas configuration (ns^2np^6).
- (iii) Larger the size of anion higher is its polarisability.

61. Arrange the following species Ag^+ , Cu^+ , K^+ in increasing order of polarising power ?
 (A) $K^+ > Cu^+ > Ag^+$ (B) $Ag^+ > K^+ > Cu^+$
 (C) $Cu^+ > Ag^+ > K^+$ (D) $Ag^+ > Cu^+ = K^+$
62. Among the following $MgCl_2$, $NaCl$, Na_2S , MgS compound having least melting point and highest melting point are respectively:
 (A) MgS & $NaCl$ (B) $NaCl$ & Na_2S
 (C) $MgCl_2$ & $NaCl$ (D) $NaCl$ & MgS
63. The correct order of polarisability I^- , Br^- , Cl^- , F^- is :
 (A) $I^- > Br^- > Cl^- > F^-$ (B) $I^- > Br^- = Cl^- > F^-$
 (C) $I^- = Br^- = Cl^- > F^-$ (D) $I^- = Br^- < Cl^- = F^-$

Passage-22

Valence shell electron-pair repulsion theory predicts the shape of a molecule by considering the most stable configuration of the bond angles in the molecule. The main points of the theory are :

- (i) Electron pairs in the valence shell of the central atom of a molecule, whether bonding or lone pairs are regarded as occupying localised orbitals. These orbitals arrange themselves in so as to minimise the mutual electronic repulsions.
- (ii) The magnitude of the different types of electronic repulsions follow the order given below: lone pair-lone pair > lone pair-bonding pair > bonding pair-bonding pair. These repulsive forces alter the bond angles of the molecule or ion.
- (iii) The electron repulsion between two pairs of electrons will be minimum if they are as far apart as possible. On this basis, the following geometrical arrangements are most suited.

64. The correct order of the bond angles is :
 (A) $NH_3 > PH_3 > AsH_3 > SbH_3$ (B) $PH_3 > AsH_3 > SbH_3 > NH_3$
 (C) $AsH_3 > SbH_3 > NH_3 > PH_3$ (D) $NH_3 > PH_3 = AsH_3 = SbH_3$
65. Which of the following statement is correct ?
- | Compound | Structure | Compound | Structure |
|------------|-------------|------------|-----------|
| (A) H_2O | Tetrahedral | (B) H_2O | Linear |
| (C) NH_3 | Tetrahedral | (D) NH_3 | Pyramidal |
66. Which of the following is correct for the bond angle .
 (A) $ClO_2 > Cl_2O$ (B) $H_2O > NH_3$ (C) $OF_2 > H_2O$ (D) $CH_4 < PH_3$

Passage-25

The valence bond theory was the first quantum mechanical theory of bonding to be developed, and can be regarded as a way of expressing Lewis concepts in terms of wave functions.

In valence bond theory, the wave function of an electron pair is formed by superimposing the wave function for the separated fragments of the molecule. And, the shape of the molecule depends on the orientation of the hybridized orbitals.

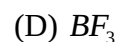
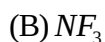
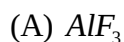
73. The maximum number of combinations of atomic orbitals that are possible in the valence shell of C-atom is
 (A) 4 (B) 3 (C) 2 (D) 1
74. Trigonal pyramidal structure is associated with 1. SO_3^{2-} 2. CO_3^{2-} 3. NO_3^-
 (A) 1 only (B) 1 & 3 only (C) 2 & 3 only (D) 1 & 2 only
75. Which one of the following is correct ?
 (A) XeF_2 and BeCl_2 are both linear with sp hybridization
 (B) XeF_2 and BeCl_2 are both linear with sp & sp^3d hybridization
 (C) XeF_2 is angular and BeCl_2 is linear with sp and sp^3d hybridization
 (D) XeF_2 and BeCl_2 both are linear with sp^3d and sp hybridization

Passage-26

Lewis concept of covalency of an element involved octet rule. Later on it was found that many elements in their compounds e.g., BeF_2 , BF_3 etc. have incomplete octet whereas PCl_5 , SF_6 etc. have expanded octet. This classical concept also failed in predicting the geometry of molecules. Modern concept of covalence was proposed in terms of valence bond theory proposed by Heitler and London and later on modified by Pauling and Slater. Hybridisation concept along with valence bond theory although successfully explained the geometry of various molecules but failed in many molecules. The geometry of such molecules was explained by VSEPR concept. Finally molecular orbital theory was proposed by Hund-Mulliken to explain many other anomalies.

76. Which are true statements among the following:
 1) I_3^+ has bent structure 2) $p\pi - d\pi$ bonds are present in SO_2
 3) SeF_4 and CH_4 has same shape 4) XeF_2 and CO_2 has same shape
 5) SF_4 is see-saw structure whereas ICl_3 is T shaped
 (A) 1, 2, 4, 5 (B) 1, 2, 3, 4 (C) 2, 3, 4, 5 (D) 1, 3, 4, 5
77. Which statements are correct
 1) Bond angle of $\text{PF}_3 > \text{PCl}_3$ 2) Bond angle of $\text{PCl}_3 < \text{PBr}_3$
 3) $(\text{CH}_3)_3\text{N}$ is pyramidal whereas $(\text{SiH}_3)_3\text{N}$ is planar but both show sp^3 hybridization.
 4) Multiple bonds also influence the geometry of molecule and thus C_2H_4 has $\angle\text{HCH} = 116^\circ$ and RCHO has $\angle\text{HCO} = 122^\circ$
 5) PCl_5 is super octet molecule
 (A) 1, 3, 4, 5 (B) 1, 2, 3, 4 (C) 2, 3, 4, 5 (D) 2, 5

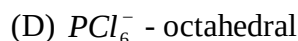
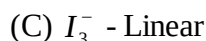
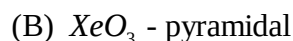
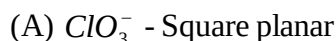
78. In which of the following molecules all the bonds are not equal



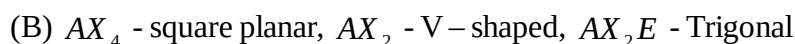
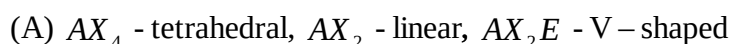
Passage-27

Sidgwick and Powell and later Gillespie and Nyholm developed valence shell electron pair repulsion theory (VSEPR) to explain the shapes of molecules. According to them a regular geometry results if the central atom has bond pairs and no lone pairs and a distorted geometry arises due to bond pairs and lone pairs around central atom. Based on number of bond pairs and lone pairs, they predicted the shapes and approximate bond angles in molecules. Now answer the following.

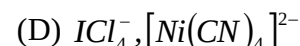
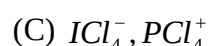
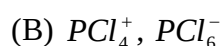
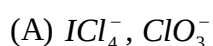
79. The incorrect matching among the following



80. VSEPR notation and probable geometry of PCl_4^+ , $HgCl_2$, $SnCl_2$ are



81. In which of the following pairs each has a square planar structure?



Passage-28

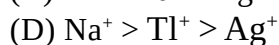
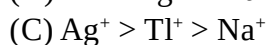
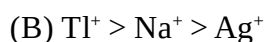
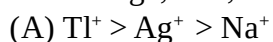
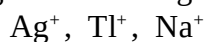
In the ionic bond, the cation tend to polarise the electron cloud of the anion by pulling electron density towards itself. This causes development of covalent character in ionic bond because the electron density gets localised in between the nuclei. The tendency of cation to bring about the polarisation of anion is expressed as its polarising power. The ability of ion to undergo polarisation is called its polarisability. The polarising power of cation and polarisability of anion are decided on the basis of Fajan rules as given below :

(i) Smaller the cation higher is its polarising power.

(ii) Cations with pseudo noble gas configuration ($ns^2np^6nd^{10}$) have relatively high polarising power than those with noble gas configuration (ns^2np^6).

(iii) Large the size of anion higher is its polarisability.

82. Arrange the following species in increasing order of polarising power ?



83. Among the following $MgCl_2$, $NaCl$, Na_2S , MgS compound having least melting point and highest solubility respectively is :



84. The correct order of decreasing polarizability of ion is :
 (A) Cl^- , Br^- , I^- , F^- (B) F^- , I^- , Br^- , Cl^-
 (C) I^- , Br^- , Cl^- , F^- (D) F^- , Cl^- , Br^- , I^-

Passage-29

Difference in polarities of bonds is expressed on a numerical scale. The polarity of a molecule is indicated in terms of **dipole moment** (μ). To measure dipole moment, a sample of the substance is placed between two electrically charged plates. Polar molecules orient themselves in the electric field causing the measured voltage between the plates to change.

The dipole moment is defined as the product of the distance separating charges of equal magnitude and opposite sign, with the magnitude of the charge. The distance between the positive and negative centres is called the bond length thus,

$$\mu = \text{electric charge} \times \text{bond length} \\ = q \times d$$

85. Which of the following molecule have zero dipole moment
 (A) PCl_5 (B) BF_3 (C) CCl_4 (D) All of these
86. The correct order of dipole moment is:
 (A) $\text{CH}_3\text{Cl} > \text{CH}_3\text{F}$ (B) $\text{HCl} > \text{HF}$ (C) $\text{NF}_3 > \text{NH}_3$ (D) $\text{NH}_3 > \text{H}_2\text{O}$
87. Which of the following option is correct ? [State T for true and F for false]
 (I) Dipole moment is a vector quantity, i.e. it has both magnitude as well as direction.
 (II) CO_2 is non polar because its dipole moment $\neq 0$
 (III) Generally in any molecules bond angle is increases dipole moment will be also increases
 (IV) We can determine percentage ionic character with the help of dipole moment
 (A) TTFF (B) TTFT (C) FFTT (D) FTTF

Passage-30

Valence shell electron-pair repulsion theory predicts the shape of a molecule by considering the most stable configuration of the bond angles in the molecule. The main points of the theory are :

- (i) Electron pairs in the valence shell of the central atom of a molecule, whether bonding or lone pairs are regarded as occupying localised orbitals. These orbitals arrange themselves in so as to minimise the mutual electronic repulsions.
- (ii) The magnitude of the different types of electronic repulsions follow the order given below: lone pair-lone pair > lone pair-bonding pair > bonding pair-bonding pair. These repulsive forces alter the bond angles of the molecule or ion.
- (iii) The electron repulsion between two pairs of electrons will be minimum if they are as far apart as possible. On this basis, the following geometrical arrangements are most suited.

88. The nitrogen atoms in NH_3 , NH_2^- and NH_4^+ are all surrounded by eight electrons. When these species are arranged in increasing order of H – N – H bond angle, correct order is :
 (A) NH_3 , NH_2^- , NH_4^+ (B) NH_4^+ , NH_2^- , NH_3
 (C) NH_3 , NH_4^+ , NH_2^- (D) NH_2^- , NH_3 , NH_4^+
89. Among the following the pair in which the two species are not isostructural is :
 (A) SiF_4 & SF_4 (B) IO_3^- & XeO_3 (C) BH_4^- & NH_4^+ (D) PF_6^- & SF_6
90. $(\text{CH}_3)_3\text{N}$ is pyramidal in shape while $(\text{SiH}_3)_3\text{N}$ is planar in shape because
 (A) $\pi\pi$ - $d\pi$ back bonding is present in $(\text{SiH}_3)_3\text{N}$
 (B) $\pi\pi$ - $d\pi$ back bonding is not present in $(\text{CH}_3)_3\text{N}$
 (C) In $(\text{SiH}_3)_3\text{N}$ lone pair e^- of nitrogen shift to vacant d-orbital of Si atom so sp^2 type hybridisation may be present and structure becomes planar.
 (D) All of these

Passage-31

Valence shell electron-pair repulsion theory predicts the shape of a molecule by considering the most stable configuration of the bond angles in the molecule. The main points of the theory are :

- (i) Electron pairs in the valence shell of the central atom of a molecule, whether bonding or lone pairs are regarded as occupying localised orbitals. These orbitals arrange themselves in so as to minimise the mutual electronic repulsions.
- (ii) The magnitude of the different types of electronic repulsions follow the order given below: lone pair-lone pair > lone pair-bonding pair > bonding pair-bonding pair. These repulsive forces alter the bond angles of the molecule or ion.
- (iii) The electron repulsion between two pairs of electrons will be minimum if they are as far apart as possible. On this basis, the following geometrical arrangements are most suited.

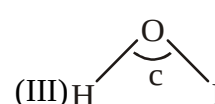
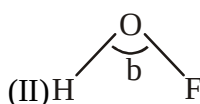
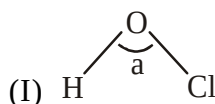
91. The nitrogen atoms in NH_3 , NH_2^- and NH_4^+ are all surrounded by eight electrons. When these species are arranged in increasing order of H – N – H bond angle, correct order is :
 (A) NH_3 , NH_2^- , NH_4^+ (B) NH_4^+ , NH_2^- , NH_3
 (C) NH_3 , NH_4^+ , NH_2^- (D) NH_2^- , NH_3 , NH_4^+
92. Select pair of compounds in which both have different hybridisation but have same molecular geometry.
 (A) BF_3 , BrF_3 (B) ICl_2^- , BeCl_2 (C) BCl_3 , PCl_3 (D) PCl_3 , NCl_3
93. Amongst NO_3^- , AsO_3^{3-} , CO_3^{2-} , ClO_3^- , SO_3^{2-} and BO_3^{3-} , Which of the following option have only planar species—
 (A) CO_3^{2-} , SO_3^{2-} , BO_3^{3-} (B) AsO_3^{3-} , ClO_3^- , SO_3^{2-}
 (C) NO_3^- , CO_3^{2-} , BO_3^{3-} (D) SO_3^{2-} , NO_3^- , BO_3^{3-}

Passage-32

Valence shell electron-pair repulsion theory predicts the shape of a molecule by considering the most stable configuration of the bond angles in the molecule. The main points of the theory are:

- (i) Electron pairs in the valence shell of the central atom of a molecule, whether bonding or lone pairs are regarded as occupying localised orbitals. These orbitals arrange themselves in so as to minimise the mutual electronic repulsions.
- (ii) The magnitude of the different types of electronic repulsions follow the order given below: lone pair-lone pair > lone pair-bonding pair > bonding pair-bonding pair. These repulsive forces alter the bond angles of the molecule or ion.
- (iii) The electron repulsion between two pairs of electrons will be minimum if they are as far apart as possible. On this basis, the following geometrical arrangements are most suited.

94. Which of the following pairs of compound will have not same structure ?
 (A) CO_2 , N_3^- (B) NO_2^- , O_3 (C) CO_3^{2-} , NO_3^- (D) N_3^- , NO_2^-
95. In the following compounds



Which of the following order is correct for bond angle ?

- (A) $a > b = c$ (B) $b > c > a$ (C) $b < a < c$ (D) $a > b > c$
96. In which of the following molecules/ions are all the bonds not equal ?
 (A) SF_4 (B) SiF_4 (C) XeF_4 (D) BF_4^-

CHEMICAL EQUILIBRIUM

Passage-33

Physical and chemical equilibria respond to stress, e.g., change in pressure, temperature and concentration of reactants and products. According to Le Chatelier's principle a system at equilibrium, when subjected to a disturbance (stress), responds in a way that to minimize the effect of the disturbance (stress), responds in a way that tends to minimize the effect of the disturbance.

Effect of pressure: The principle implies that if a system at equilibrium is compressed, the reaction will adjust so as to minimize the increase in pressure. Thus, if volume of gas phase reaction system is reduced, the equilibrium shifts in the direction in which number of moles decreases. Although equilibrium constant is independent of pressure (at a fixed temperature), the various concentrations are changed on reducing the volume of the system, i.e. increasing the pressure. In case of physical equilibria on increasing pressure the system shifts in the direction of species of higher density as to reduce the effect of increased pressure.

However, if increase in pressure effected by introducing an inert gas to reaction vessel of constant volume, partial pressures or concentrations of various species remain unchanged as they continue to occupy the same volume. But at constant pressure condition, introduction of inert gas into the reaction system will lead to increase in volume and in consequence to the change in concentrations of various species. Under this condition, the equilibrium shifts in the direction in which number of moles of gaseous species increases.

Effect of temperature: Le-Chatelier's principle predicts a system at equilibrium will tend to shift in the endothermic direction when temperature is raised, for then energy is absorbed as heat and the rise in temperature is opposed. Conversely, an equilibrium will shift in the exothermic direction if the temperature is lowered, for then that energy is released and the reduction in temperature is opposed. Vant Hoff equation shows the dependence of equilibrium constant K on temperature as:

$$\frac{d}{dT} \ln K = \frac{\Delta H^\circ}{RT^2} \text{ or } \ln K = \text{constant} - \frac{\Delta H^\circ}{R} \cdot \frac{1}{T}.$$

Effect of addition of reactants or products: on addition of one or more reactants the equilibrium will shift to the products causes the equilibrium to shift in backward direction..

97. Consider the equilibrium, $2\text{CO(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO}_2\text{(g)} + \text{heat}$
If O_2 is added and volume of the reaction vessel is reduced, the equilibrium will
(A) shift in forward direction (B) Shift in reverse direction
(C) remains unchanged (D) be unpredictable
98. The volume of the systems describing the equilibria
(I) $\text{heat} + \text{MgCO}_3\text{(s)} \rightleftharpoons \text{MgO(s)} + \text{CO}_2\text{(g)}$
(II) $2\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO(g)} + \text{heat}$
is decreased. The equilibrium will shift
(A) to the right in both (i) and (II)
(B) to the left in both (I) and (II)
(C) to the left in (I) and to the right in (II)
(D) to the right in (I) and to the left in (II)
99. To the system $\text{X(s)} + 2\text{A(g)} + \text{heat} \rightleftharpoons 2\text{Y(s)} + 3\text{B(g)}$ at equilibrium more A(g) is added with temperature volume constant. If the new steady pressure of A(g) becomes 2.828 times of the previous steady pressure, then the no. of moles of B(g) are increased by a factor of
(A) $3^{1/3}$ (B) 3 (C) $\sqrt{2}$ (D) 2

Passage-34

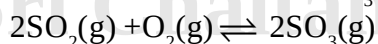
When 306 g of NH_4HS is introduced in a 4.1 litre evacuated flask and heated to 227°C then solid NH_4HS decomposes into gaseous ammonia and hydrogen sulphide. At equilibrium, it is found that 170 gm of H_2S are present in the container.

100. Mass of solid remaining in the container is -
 (A) 51 g (B) 136 g (C) 68 g (D) 100 g
101. If after establishment of equilibrium, H_2S is added in the container in such a way that partial pressure of H_2S now equals to original total pressure, then partial pressure of NH_3 at new equilibrium is -
 (A) 50 atm (B) 25 atm (C) 100 atm (D) 75 atm
102. If at 227°C , ammonia also starts decomposing into nitrogen and hydrogen and establishes equilibrium ($2\text{NH}_3 \rightleftharpoons \text{N}_2 + 3\text{H}_2$) then K_p for the reaction $\text{NH}_4\text{HS(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{H}_2\text{S(g)}$ and total pressure developed in the container will -
 (A) decreases, increases (B) increases, decreases
 (C) no change, no change (D) no change, increases

Passage-35

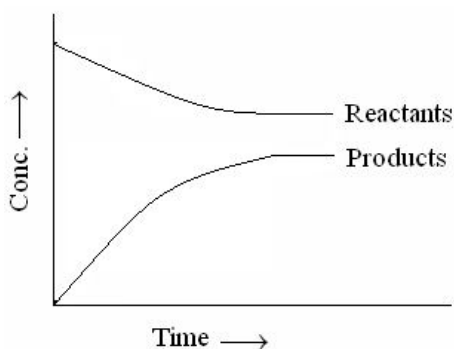
In a reversible chemical reaction, the rate of forward reaction decreases and that of backward reaction increases with the passage of time; at equilibrium the rate of forward and backward reactions become same.

Let us consider the formation of SO_3 in the following reversible reaction:

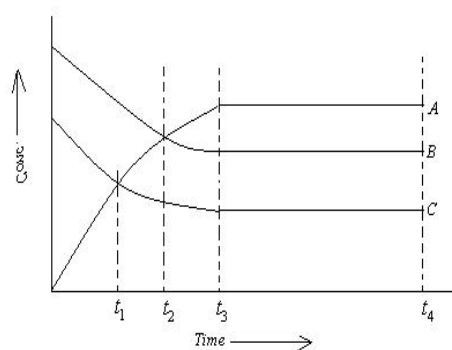


Following graphs are plotted for this reaction:

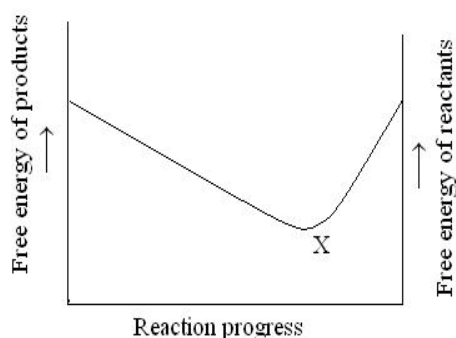
Graph - 1



Graph - 2



Graph - 3



103. In the graph(1), the equilibrium state is reached at:
 (A) t_1 (B) t_2 (C) t_3 (D) t_4

104. The graph(2) tells us that:
 (A) equilibrium is never achievable
 (B) equilibrium is achieved after the concentrations of reactants and products become equal
 (C) equilibrium is achieved before the concentrations of reactants and products become equal
 (D) none of these
105. From the graph(3), it can be interpreted that:
 (A) equilibrium is achieved at X
 (B) reaction is nearer to the completion
 (C) $\Delta G=0$ at X
 (D) all of the above

Passage-36

Physical and chemical equilibria respond to stress, e.g., change in pressure, temperature and concentration of reactants and products. According to Le Chatelier's principle a system at equilibrium, when subjected to a disturbance (stress), responds in a way that to minimize the effect of the disturbance (stress), responds in a way that tends to minimize the effect of the disturbance.

Effect of pressure: The principle implies that if a system at equilibrium is compressed, the reaction will adjust so as to minimize the increase in pressure. Thus, if volume of gas phase reaction system is reduced, the equilibrium shifts in the direction in which number of moles decreases. Although equilibrium constant is independent of pressure (at a fixed temperature), the various concentrations are changed on reducing the volume of the system, i.e. increasing the pressure. In case of physical equilibria on increasing pressure the system shifts in the direction of species of higher density as to reduce the effect of increased pressure.

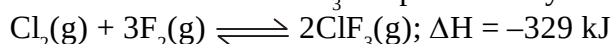
However, if increase in pressure effected by introducing an inert gas to reaction vessel of constant volume, partial pressures or concentrations of various species remain unchanged as they continue to occupy the same volume. But at constant pressure condition, introduction of inert gas into the reaction system will lead to increase in volume and in consequence to the change in concentrations of various species. Under this condition, the equilibrium shifts in the direction in which number of moles of gaseous species increases.

Effect of temperature: Le-Chatelier's principle predicts a system at equilibrium will tend to shift in the endothermic direction when temperature is raised, for then energy is absorbed as heat and the rise in temperature is opposed. Conversely, an equilibrium will shift in the exothermic direction if the temperature is lowered, for then that energy is released and the reduction in temperature is opposed. Vant Hoff equation shows the dependence of equilibrium constant K on temperature as:

$$\frac{d}{dT} \ln K = \frac{\Delta H^\circ}{RT^2} \text{ or } \ln K = \text{constant} - \frac{\Delta H^\circ}{R} \cdot \frac{1}{T}.$$

Effect of addition of reactants or products: on addition of one or more reactants the equilibrium will shift to the products causes the equilibrium to shift in backward direction..

106. The exothermic formation of ClF_3 is represented by the equation:

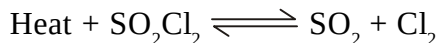


Which of the following will increase the quantity of ClF_3 in equilibrium mixture of Cl_2 , F_2 and ClF_3 ?

- (A) Increasing the temperature
 (B) Removing Cl_2
 (C) Increasing the volume of container
 (D) Increasing pressure

107. In a system: $A(s) \rightleftharpoons 2B(g) + 3C(g)$. If the concentration of C at equilibrium is increased by a factor 2, it will cause the equilibrium concentration of B to change to:
- (A) Two times of its original value (B) One half of its original value
- (C) $2\sqrt{2}$ times of its original value (D) $\frac{1}{2\sqrt{2}}$ times of its original value

108. A system in equilibrium is described by the gaseous phase equation



which of the following statements is true?

- (A) Addition of Cl_2 will shift the equilibrium towards right and temperature is raised
- (B) Addition of Cl_2 will shift the equilibrium towards left and temperature is raised
- (C) Withdrawal of Cl_2 and SO_2 will shift the equilibrium towards left and temperature is raised
- (D) Withdrawal of SO_2 and Cl_2 will shift the equilibrium towards left and temperature is lowered

Passage-37

Law of mass action can also be applied to the study of equilibria in which the substances are not in the same phase. Considering the decomposition of solid NaHCO_3 to produce solid Na_2CO_3 , gaseous CO_2 and H_2O .



Applying the law of mass action.

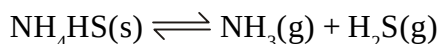
$$K'_C = \frac{[\text{Na}_2\text{CO}_3][\text{CO}_2][\text{H}_2\text{O}]}{[\text{NaHCO}_3]^2}$$

It involves two pure solids, Na_2CO_3 and NaHCO_3 . It is customary not to include the concentrations of pure solids in equilibrium expressions.

$$\frac{K'_C [\text{NaHCO}_3]^2}{[\text{Na}_2\text{CO}_3]} = [\text{CO}_2][\text{H}_2\text{O}] \quad \text{or} \quad K_C = [\text{CO}_2][\text{H}_2\text{O}]$$

In terms of partial pressures we have $K_p = P_{\text{CO}_2} \times P_{\text{H}_2\text{O}}$. Similarly, in reactions in which a reactant or product occurs as a pure liquid phase, the concentration of that substance in the pure liquid is also constant. As a result, the concentrations of pure solid and pure liquid phases do not appear in the equilibrium constant expression.

109. Ammonium hydrogen sulphide dissociates according to the equation



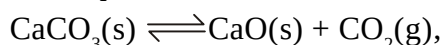
If the observed pressure at equilibrium is 2280 torr atm at 380 K, what is the equilibrium constant K_p of the reaction?

- (A) 1.5 (B) 2.25 (C) 3 (D) 4.5

110. A vessel at 1000 K contains carbon dioxide with a pressure of 0.5 atm. Some of the carbon dioxide is converted to carbon monoxide on addition of graphite. Calculate the value of K_p if total pressure at equilibrium is 0.8 atm.

- (A) 1.8 atm (B) 0.8 atm (C) 3.6 atm (D) 4 atm

111. A sample of $\text{CaCO}_3(s)$ is introduced into a sealed container of volume 0.64 litre and heated to 1000 K until equilibrium is reached. The equilibrium constant for the reaction,



is 4×10^{-2} atm at this temperature. Calculate the mass of CaO (in mg) present at equilibrium.

- (A) 3.584 (B) 1.792 (C) 17.92 (D) 35.84

Passage-38

10 mole ammonia is heated at 15atm from 27°C to 347°C assuming volume constant . The pressure at equilibrium is found to be 50atm. The equilibrium constant for dissociation of NH_3 :



112. The degree of dissociation of NH_3 is
 (A) 61.3% (B) 20% (C) 48% (D) 96%
113. The volume of the container in which the gas is heated is
 (A) 16.42L (B) 8.21L (C) 20 L (D) 15 L
- 114.. The vapor density of the mixture at equilibrium is
 (A) 6.97L (B) 24.82 (C) 14.14 (D) 5.27

COORDINATION COMPOUND

Passage-39

Because of the presence of vacant d-orbitals ,transition metals can accept ligands. Depending on the strengths of the ligands the complexes have different hybridization of the central atoms and consequently exhibit different magnetic properties. Thus strong ligands like CN^- , CO , NH_3 , NO_2^- etc. cause maximum pairing of electrons and hence are diamagnetic in have low spins. Weak ligands like F^- , Cl^- , Br^- , I^- etc. produce weak ligand field resulting in a large number of unpaired electrons in the complex. They are referred to as high spin complexes.

- 115 They hybridization in the complexes $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ are respectively
 (A) sp^3, sp^3 (B) sp^3, dsp^2 (C) dsp^2, sp^3 (D) dsp^2, dsp^2
- 116 The species that is paramagnetic is
 (A) $[\text{Fe}(\text{CN})_6]^{4-}$ (B) $[\text{Cr}(\text{CN})_6]^{-3}$ (C) $[\text{Ni}(\text{CO})_4]$ (D) $[\text{Co}(\text{CN})_6]^{-3}$
117. Three moles of AgNO_3 solution is added to 1 mole of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ and mixed thoroughly. It was found that 1 mole AgNO_3 remained in solution. The formula of the complex can be represented as
 (A) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (B) $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$
 (C) $[\text{CrCl}_3(\text{H}_2\text{O})_3]3\text{H}_2\text{O}$ (D) $[\text{CrCl}_2(\text{H}_2\text{O})_6]\text{Cl} \cdot 2\text{H}_2\text{O}$

Passage-40

A, B, and C are three complexes of chromium (III) with the empirical formula $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. All the three complexes have water and chloride ions as ligands . Complex A does not react with concentrated H_2SO_4 , whereas complexes B and C loss 6.75% and 13.5% of their original weight respectively, on treatment with concentrated H_2SO_4 .

118. Complex A is :
 (A) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (B) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (C) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ (D) none of these

119. Which of the following statement is correct ? [State T for true and F for false]

- (i) The effective atomic number of the complex B is 33
 (ii) H_2SO_4 is act in above question is used as a dehydrating agent
 (iii) The number of ions provided by complex B is 3
 (A) TTT (B) FTF (C) TTF (D) FFF

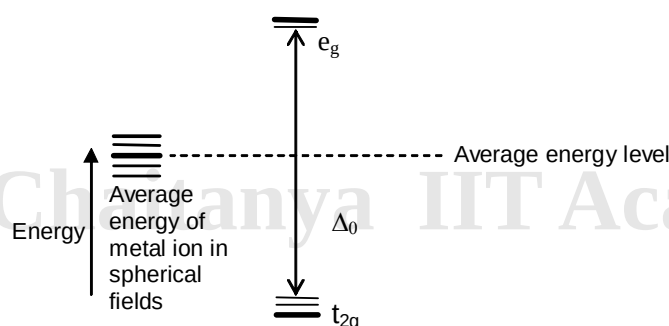
120. In the above complexes, which of the following complex is maximum hydrated ?

- (A) Complex C (B) Complex B (C) Complex A
 (D) All are equally hydrated

Passage-41

In octahedral complexes the filling of t_{2g} orbitals decreases the energy of a complex, that is makes it more stable by $-0.4\Delta_0$ per electron. Filling the e_g orbitals increases the energy by $+0.6\Delta_0$ per electron. The total crystal field stabilisation energy is given by

$$\text{CFSE (octahedral)} = -0.4\Delta_0(t_{2g}) + 0.6\Delta_0(e_g)$$



The CFSE increases the thermodynamic stability of the complexes i.e it affects the actual lattice energy over the theoretically calculated energy that does not take CFSE into account. In studying tetrahedral complexes a regular tetrahedron is related to a cube. One atom is at the centre and four of eight corners are occupied by the ligands. The directions of x, y, z point to the centres of the faces of the cube. The directions of the approach of the ligands does not coincide exactly with the atomic orbitals.

In these complexes the e_g orbitals are more stable compared to the t_{2g} orbitals.

121. In an octahedral complex X the central metal ion has 24 electrons. If its pairing energy is $+0.8\Delta_0$ (where Δ_0 is the difference in energy between the t_{2g} and the e_g orbitals). Will it obey Hund's rule and what is its spin magnetic moment
 (A) Yes, 2.83 (B) No, 2.83 (C) Yes, 4.89 (D) No, 4.89
122. During the formation of a tetrahedral complex the angle between the e_g orbital and the central metal is X° and that of t_{2g} orbital and the central metal is Y° then $X^\circ - Y^\circ$ is.
 (A) 24.3° (B) 19.3° (C) 35.36° (D) 21.24°
123. Considering Mn^{2+} if the measured lattice energy is X kJ/mol and the calculated lattice energy is Y kJ/mol then their relation between X and Y in a strong and a weak ligand field respectively is
 (A) $X > Y$, $X = Y$ (B) $X > Y$, $X < Y$ (C) $X < Y$, $X > Y$ (D) $X < Y$, $X = Y$

Passage-42

Compounds containing M-C bonds are called organometallic compounds of which carbonyls constitute an important part. Synergic effect in bonding in the metal carbonyls makes them stable species, metal carbonyls abide by the effective atomic number (EAN) rule.

124. Which of the following is not an organometallic compound ?

- (A) $Al(C_2H_5O)_3$ (B) $Fr(C_5H_5)_2$ (C) $[PtCl_3(C_2H_2)]^-$ (D) $Cr(CO)_6$

125. The value of x in $[Mn(CO)_5]^{x-}$ is

- (A) 3 (B) 2 (C) 1 (D) 0

126. Which of the following is true for the synergic bonding in metal carbonyls ?

- (A) The C-O bond length increases (B) The bond order of CO increases
(C) The C-O bond length does not change (D) None of these

Passage-43

Valence bond theory explains the magnetic behaviour of coordination species. Depending upon the magnetic behaviour one can understand whether the inner d-orbitals or outer d-orbitals are being used in a particular hybridisation. However, the colours of coordination species can be explained by the crystal field theory.

127. Which of the following compounds is diamagnetic as well as coloured?

- (A) $[Fe(CN)_6]^{4-}$ (B) $[Ni(CN)_4]^{2-}$ (C) $[Co(NH_3)_6]^{3+}$ (D) All of these

128. Though NH_3 is a strong field ligand, in which of the following compounds, the d-electrons are not paired ?

- (A) $[Cr(NH_3)_6]^{3+}$ (B) $[Ni(NH_3)_6]^{2+}$ (C) $[CrCl_3(NH_3)_3]$ (D) All of these

129. The correct order of paramagnetic moment is given by

- (A) $[Cu(NH_3)_4]^{2+} < [FeF_6]^{3-} < [Cr(NH_3)_6]^{3+}$
(B) $[FeF_6]^{3-} < [Cu(NH_3)_4]^{2+} < [Cr(NH_3)_6]^{3+}$
(C) $[Cu(NH_3)_4]^{2+} < [Cr(NH_3)_6]^{3+} < [FeF_6]^{3-}$
(D) $[Cr(NH_3)_6]^{3+} < [Cu(NH_3)_4]^{2+} < [FeF_6]^{3-}$

Passage-44

The Carbon-metal bond in carbonyls may be represented as $M \leftarrow C \equiv O$. A stronger second bond is formed by back bonding. This arises from sideways overlap of π^* 2Py orbital of carbon with full dxy orbital of the metal.

130. Because of back bonding 'CO' is called as:

- (A) π - acceptor ligand (B) π - donor ligand
(C) σ - acceptor base (D) σ - donor ligand

131. In metal carbonyls the C-O bond length

- (A) Decreases from double bond length to triple length.
(B) Increases from triple bond length to double bond length.
(C) Remains as triple bond length. (D) Cannot be predicted.

132. Which pair of carbonyl is correctly matched with shape.

- (A) Nickel carbonyl – Square planar
- (B) Iron carbonyl – Pentagonal bipyramidal
- (C) Vanadium carbonyl – Tetrahedral
- (D) Chromium carbonyl – Octahedral

Passage-45

Coordination compounds can also exhibit ionization and linkage isomerism types along with other isomerism types. In ionization isomerism type, the exchange of ions between coordination sphere and ionic sphere is observed. On the other hand, is linkage isomerism, the change in the donor atom of a ligand is observed.

133. Which one of the following is correct with respect to SCN^- ion ?

- (A) metal - S bonds are stable in solvent with low dielectric constant
- (B) metal - N bonds are stable in solvent with high dielectric constant
- (C) metal - S bonds are stable in solvent with high dielectric constant
- (D) in any solvent metal - S and metal - N bonds are stable

134. Which one of the following is not an example for ionization isomerism:

- (A) $[(NH_3)_5Co^{18}OH] + NO_2 \rightarrow [(NH_3)_5Co(NO_2)] + ^{18}OH$
- (B) $[(NH_3)_4Co(H_2O)Cl] + 2Br \rightarrow [(NH_3)_4Co(Br_2)] + Cl \cdot H_2O$
- (C) $[(NH_3)_5Co(SO_4)] + NO_3 \rightarrow [(NH_3)_5Co(NO_3)] + SO_4$
- (D) $[(NH_3)_4Co(NO_2)Cl] + Cl \rightarrow [(NH_3)_4CoCl_2] + NO_2$

135. How many ways are possible to represent a coordination complex with a molecular formula

$[M^{m\pm}a_4b_2]^{n\pm}$ (when m may or may not be equal to n) in - terms of geometrical isomerism

- (A) 12 Cis forms and 3 trans forms
- (B) 6 Cis forms and 6 trans forms
- (C) 4 Cis forms and 4 trans forms
- (D) 8 Cis and 4 trans forms

Passage-46

A metal complex having composition $Cr(NH_3)_4Br_2I$ was isolated in two forms (x) and (y). Form (x) reacts with $AgNO_3$ to give a pale yellow precipitate which is partially soluble in excess of NH_4OH . Whereas (y) gives a greenish yellow precipitate which is insoluble in NH_4OH .

136. (A) The formula of (x) and (y) are $[Cr(NH_3)_4IBr]Br$ and $[Cr(NH_3)_3Br_2I]NH_3$
 (B) The formula of (x) and (y) are $[Cr(NH_3)_4IBr]Br$ and $[Cr(NH_3)_4Br_2]I$ respectively
 (C) The formula of (x) and (y) are both $[Cr(NH_3)_4Br]BrI$
 (D) The formula of (x) and (y) are $[Cr(NH_3)_2IBr_2](NH_3)_2$
137. Both the (x) form and (y) form show
 (A) linkage isomerism
 (B) Co-ordination isomerism
 (C) ionization isomerism
 (D) none of these

138. The oxidation state and co-ordination number of the central atom in the complex is

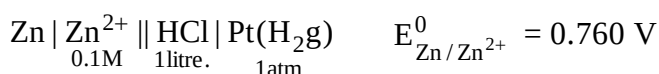
- (A) +6, 6
- (B) +6, 5
- (C) +3, 5
- (D) +3, 6

ELECTRO CHEMISTRY

Passage-47

The potential associated with each electrode is known as electrode potential. If the concentration of each species taking part in the electrode reaction is unit (if any gas appears in the electrode reaction, it is confined to 1 atmospheric pressure) and further the reaction is carried out at 298 K, then the potential of each electrode is said to be the standard electrode potential. By convention, the standard electrode potential of hydrogen electrode is 0.0 volt. The electrode potential value for each electrode process is a measure of relative tendency of the active species in the process to remain in the oxidised/reduced form. A negative E° means that the redox couple is a stronger reducing agent than the H^+/H_2 couple. A positive E° means that the redox couple is a weaker reducing agent than the H^+/H_2 couple. The metal with greater positive value of standard reduction potential forms the oxide of greater thermal stability.

139. Calculate the minimum mass of NaOH required to be added in R.H.S. to consume all the H^+ present in R.H.S. of the cell of emf + 0.701 volt at 25°C before its use.



- (A) 1.264 g (B) 4 g (C) 3.2 g (D) 2.258

140. The emf of the cell, $Ag | AgI \text{ in } KI_{0.05M} | NH_4NO_3 \text{ Sol. } | AgNO_3_{0.05M} | Ag$

is 0.788 volt at 25 °C . The degree of dissociation of KI and silver nitrate in the above solution is 0.90 each . Calculate the solubility product of AgI

- (A) 9.472×10^{-17} (B) 4×10^{-17} (C) 11×10^{-17} (D) 2.105×10^{-17}

141. The standard reduction potential at 25°C of the reaction $2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$ is – 0.8277 volt. Calculate the equilibrium constant for the reaction in which water is forming from its ions at 25°C.

- (A) 10^{-14} (B) 10^{-7} (C) 10^{14} (D) 10^7

Passage-48

Kohlrausch Law of independent migration of ions relates the ionic conductivities and molar conductivities of the electrolyte. Mobility of ion is not just the speed but, speed under a potential gradient. Moreover mobility of an ion is directly proportional to current carried by it. The fraction of total current carried by an ion is the transport number.

Given ionic mobilities of NH_4^+ , H^+ , ClO_4^- , Na^+ are 6.6×10^{-4} , 3.62×10^{-3} , 5.7×10^{-4} , 5.2×10^{-4} , $cm^2 \text{ Volt}^{-1} \text{ Sec}^{-1}$ respectively at infinite dilution.

142. The equivalent conductance of $NaClO_4$ in $S \text{ cm}^2 \text{ eq}^{-1}$ is

- (A) 105.18 (B) 118.7 (C) 404.3 (D) 356.8

143. Which of the following incorrect with respect to Kohlrausch law

- (A) $\Lambda^\circ Al_2(SO_4)_3 = 3\lambda^\circ_{(Al^{+3})} + 2\lambda^\circ_{(SO_4^{-2})}$ (B) $\Lambda^\circ MgCl_2 = \lambda^\circ_{(Mg^{+2})} + 2\lambda^\circ_{(Cl^-)}$
(C) $\Lambda^\circ NH_4NO_3 = \lambda^\circ_{(NH_4^+)} + \lambda^\circ_{(NO_3^-)}$ (D) $\Lambda^\circ FeCl_3 = \lambda^\circ_{(Fe^{+3})} + 3\lambda^\circ_{(Cl^-)}$

144. The ionic mobility of Na^+ if a potential of 15 Volt is applied across the electrodes 5cm apart.

- (A) 5.2×10^{-4} (B) 15.6×10^4 (C) 10.4×10^{-4} (D) 20.8×10^{-4}

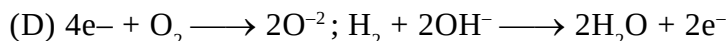
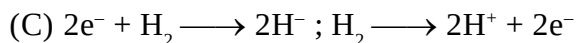
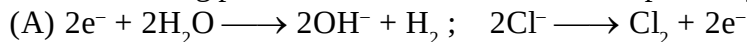
Passage-49

An aqueous solution of NaCl on electrolysis gives $H_2(g)$, $Cl_2(g)$ and NaOH according to reaction
 $2Cl^-(aq) + 2H_2O \longrightarrow 2OH^-(aq) + H_2(g) + Cl_2(g)$

A direct current of 25 ampere with a current efficiency of 62% is passed through 20 litre of NaCl solution (20% by weight)

Answer the following questions based on above passage.

145. Reaction taking place at cathode and at anode respectively are:



146. Time required to produce 1 kg of Cl_2 is –

(A) 30.2 hr

(B) 15.1 hr

(C) 24.35 hr

(D) 48.71 hr

147. Molarity of solution with respect to OH^- ions is –

(A) 2.8

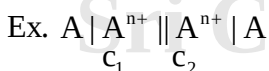
(B) 0.35

(C) 1.4

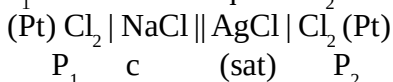
(D) 0.7

Passage-50

Concentration cell is one in which anode and cathode of a voltaic cell is made of same electrode



c_1 should not be equal to c_2 to have some emf in the cell. Now consider the cell



K_{sp} of AgCl is 10^{-12}

148. If $P_1 = P_2$, required condition so that the above cell has positive emf

(A) $c > 10^{-6} M$

(B) $c < 10^{-6} M$

(C) $P_1 = P_2 = 1 atm$

(D) Emf of the cell cannot be found as it is not a concentration cell

149. If $c = 10^{-6}$ which of the following conditions will give positive emf

(A) $P_1 > P_2$

(B) $P_1 < P_2$

(C) $P_1 = P_2$

(D) None

150. Which one of the statements is correct about the cell (Pt) $H_2 | HCl || NaOH | H_2$ (Pt)

$1 atm \quad c \quad c \quad 1 atm$

(A) It is not a concentration cell

(B) It has a positive emf

(C) It has a negative emf

(D) cannot say anything about emf without knowing the value of c

GASEOUS STATE

Passage-51

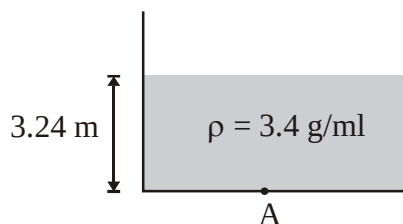
The pressure of the gas is the force exerted by the gas per unit area of the walls of the container. The pressure of gases is measured by a device known as manometer. Two types of manometers, open end manometer and closed end manometer, are commonly used to measure gaseous pressure. Pressure of one atmosphere (1 atm) is defined as the pressure that can support a column of mercury of 76 cm height at 0°C (density of mercury = 13.5951 g cm⁻³) and at standard gravity = 980.665 cm s⁻². One atmosphere is also referred to as 760 torr.

$$1 \text{ atm} = 76.0 \text{ cm of mercury} = 760 \text{ mm of mercury} \\ = 760 \text{ torr}$$

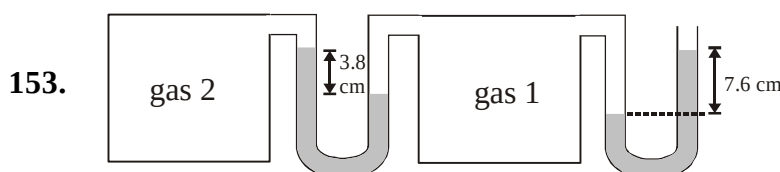
$$1 \text{ atm} = 101.325 \times 10^3 \text{ N m}^{-2} = 101.325 \text{ kPa}$$

An older unit of pressure is 'bar' which is equal to 10⁵ Pa.

151. At point A calculate pressure due to column of liquid in atm.



- (A) 1 atm (B) 2 atm (C) 1.066 atm (D) .81 atm
152. The total pressure at point A is equal to x m height of water column then x is:
 (A) 10.33 m (B) 21.33 m (C) 2.133 m (D) 11.02 m



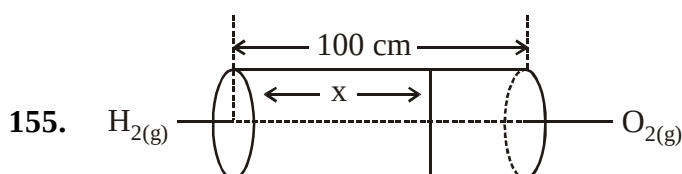
153. In above arrangement calculate pressure of gas 1 and gas 2 in atm.
 (A) 1.1, 1.05 atm (B) 1.2, 1.1 atm (C) 1, 1.1 atm (D) 1.05, 1.1 atm

Passage-52

When a gas is made to pass through a fine hole made in the wall of the container under a difference of pressure, it is known as effusion. The rate of effusion depends on the density, the temperature of the gas and the pressure gradient. Graham's law states that at a constant temperature and for constant pressure gradient the rates of effusion or diffusion of different gases are inversely proportional to the square root of their densities or molecular masses.

This law is very useful for calculating molecular weight, density, etc.... of gases. However, it should be noted that the law is true only for gases diffusing under low pressure gradient.

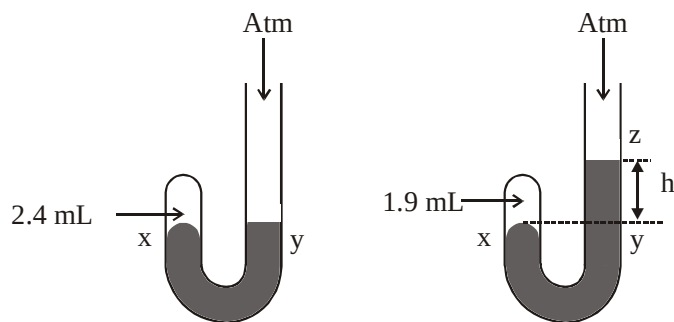
154. Rate of diffusion of hydrogen gas is $\sqrt{20}$ times rate of diffusion of a sample containing O₂ and O₃. Calculate the percentage (by moles) of O₂ in the sample.
 (A) 75 % (B) 25 % (C) 50 % (D) 90 %



155. Value of x at which first drop of H₂O is formed :
 (A) 20 cm (B) 80 cm (C) 50 cm (D) 60 cm

156. Given J-tube has 2.4 mL of air at a pressure of 1 atm. On adding mercury, volume of air is reduced to 1.9 mL as shown. Difference in the level of mercury in two columns is:

Gaseous



- (A) 700 mm (B) 200 mm (C) 900 mm (D) 760 mm

Passage-53

Ideal gas equation is represented as $PV = nRT$. Gases present in universe were found ideal in the Boyle's temperature range only and deviated more from ideal gas behaviour at high pressure and low temperature. The deviations are explained in terms of compressibility factor Z . For ideal behaviour

$Z = \frac{PV}{nRT} = 1$. The main cause to show deviations were due to wrong assumptions made about forces of attractions (which becomes significant at high pressure) and volume occupied by molecules V in ($PV = nRT$). If volume occupied by gaseous molecule is not negligible, then the term V should be replaced by the ideal volume which is available for free motion of each molecules of gas i.e., for 1 mole gas.

$$V_{actual} = \text{volume of container} - \text{volume occupied by molecules} \\ = V - b$$

where b represents the effective volume of co-volume or excluded volume occupied by molecules present in one mole of gas. Similarly, for n mole gas, ($V_{actual} = V - nb$)

The excluded volume can be calculated by considering bimolecular collisions. The excluded is the volume occupied by the sphere of $2r$ for each pair of molecule.

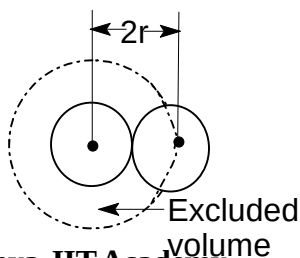
$$\text{Thus, excluded volume for one pair of molecules} = \frac{4}{3} \pi (2r)^3 = \frac{4 \times 8\pi r^3}{3} \dots\dots(i)$$

$$\therefore \text{Excluded volume for 1 molecule} = \frac{2}{3} \times 8\pi r^3 \dots\dots\dots(ii)$$

$$= 4 \times \left(\frac{4}{3} \pi r^3 \right)$$

$$= 4 \times \text{volume of one molecule} = 4V$$

$$\therefore \text{Excluded volume for } N \text{ molecules} = 4NV$$



157. Which of the following statements are correct?

- I) Larger is the value of $\frac{T_c}{P_c}$ for a gas larger would be its excluded volume
- II) Larger is the excluded volume of gas, more will be its critical volume
- III) The slope for an isochore obtained for a gas showing $P(V-b) = RT$ is $\left[\frac{R}{V-b} \right]$
- IV) The excluded volume for He is more than H_2 .
- (A) I, II, III (B) I, II, III, IV (C) II, III, IV (D) III, IV

158. Which of the following statements are correct?

- I) Rise in compressibility factor Z with increase in pressure is due to 'a'
- II) Rise in compressibility factor Z with increase in pressure is due to 'b'
- III) Ideal gas do not exist but is a useful concept

IV) For 1 mole of a van der Waals's gas, $Z = 1 + \frac{bP}{RT} - \frac{a}{RTV} + \frac{ab}{RTV^2}$

- (A) I, II, III, IV (B) II, III, IV (C) I, III, IV (D) I, IV

159. The ratio of coefficient of thermal expansion $\alpha = \frac{\left(\frac{\partial V}{\partial T} \right)_P}{V}$ and the isothermal compressibility

$K = \frac{\left(\frac{\partial V}{\partial P} \right)_T}{V}$ for an ideal gas is:

- (A) $-\frac{P}{T}$ (B) $\frac{P}{T}$ (C) $\frac{T}{P}$ (D) $-\frac{T}{P}$

Passage-54

Average K.E per mole does not depend on the nature of the gas but depends only on temperature

$$\text{Average K.E per molecule} = \frac{\text{Average K.E / mole}}{N}$$

$$= \frac{3RT}{2N} = \frac{3}{2}KT$$

Where K = Boltzmann constant

160. Which of the following expressions correctly represents the relationship between average molar kinetic energies of CO and N_2 molecules at the same temp?

- (A) $KE(CO) = K.E(N_2)$ (B) $KE(CO) > KE(N_2)$
- (C) $KE(CO) < KE(N_2)$
- (D) cannot be predicted unless the volumes of the gases are given

161. The average K.e (in joule) of the molecules in 8gm of methane at $27^\circ C$ is

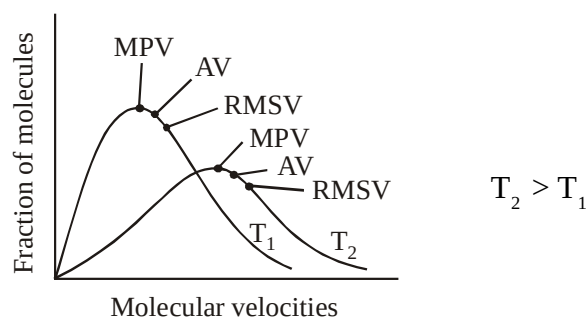
- (A) $62.14 \times 10^{-27} J$ (B) $72.68 \times 10^{-21} J$ (C) $68.2 \times 10^{-21} J$ (D) $62.14 \times 10^{-29} J$

162. Which of the following is valid at absolute zero?

- (A) K.E of the gas becomes zero but the molecular motion does not become zero
- (B) K.E and molecular motion becomes zero
- (C) K.E of the gas decreases but does not become zero
- (D) none

In gases the distribution of molecular velocities depends on temperature. The curve depicting the molecular velocities at a particular temperature vs the fraction of molecules becomes flatter and broader at higher temperature. The root mean square velocity (RMSV), average velocity (AV) and most probable velocity (MPV) at a given temperature are in the ratio

$$\text{RMSV} : \text{AV} : \text{MPV} : 1 : 0.9213 : 0.816$$



- 163 The root mean square velocity of a gas at 27°C is ' v '. At 127°C , the new root mean square velocity would be

(A) $\frac{2}{\sqrt{3}}v$ (B) $\sqrt{\frac{2}{3}}v$ (C) $\sqrt{\frac{3}{2}}v$ (D) $\frac{3}{\sqrt{2}}v$

- 164 There are two gases A and B. Ratio of rms speed of both the gases at 27°C to 127°C is 0.87 hence gas A and B could be:

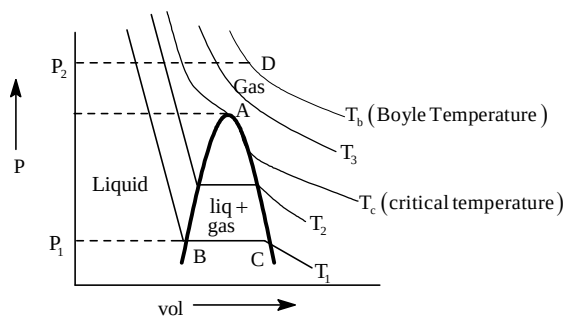
(A) O_2, SO_2 (B) N_2, CO (C) $\text{N}_2 \text{ \& } \text{CO}_2$ (D) $\text{CO}_2 \text{ \& } \text{SO}_2$

- 165 At what temperature is the RMS speed of H_2 molecules the same as that of oxygen molecules at 1327°C ?

(A) 173 K (B) 100 K (C) 400 K (D) 523 K

Passage-56

Andrews isotherm for CO_2 may be plotted as shown



166. Pressure P_1 along the line BC is called

(A) pressure of gas at T_1 (B) critical pressure
(C) vapour pressure of liquid at T_1 (D) pressure of gas below T_c

167. Which of the following is a correct statement?

P: within the area BAC, state of matter is called "vapour"

Q: At T_b , PV remains a constant

R: At point "A", liquid state of matter loses its meniscus

(A) only P,Q (B) only Q,R (C) P,Q and R (D) only P,R

168. At point 'A', which of the following is true?

P: compressibility factor becomes 0.375

Q: gas is called vander Waals' gas

R: below 'A' only one variable is required for liquefaction

(A) only P,Q (B) only Q,R (C) P,Q and R (D) only P,R

IONIC EQUILIBRIUM

Passage-57

A solution of weak acid HA was titrated with base NaOH. The equivalence point was reached when 20 ml of 0.1 M NaOH has been added. K_a of HA = 10^{-5} .

- 169 pH when 10 ml of 0.1 M HCl is added in the titrated solution :
 (A) 3 (B) 7 (C) 5 (D) 10
- 170 pH of the solution obtained by mixing 10 ml 0.2 M NaOH and 10 ml of 0.2 M HA .
 (A) 7 (B) 9 (C) 11 (D) 5
171. A solution containing 0.1M M^+ ion and 0.1 M HA , if K_{sp} of MOH is 10^{-10} then precipitate of ion will
 (A) form (B) not form (C) cannot predict (D) none of these

Passage-58

Let AB be the weak electrolyte. Considering its dissociation, $AB \rightleftharpoons A^+ + B^-$ and applying

law of mass action we have $K = \frac{[A^+][B^-]}{[AB]}$

The equilibrium constant, K, has a definite value at any given temperature. If now another electrolyte furnishing of A^+ and B^- ions be added to the above solution, it will increase the concentration of either A^+ ions or B^- ions (whichever has been added) and in order that K may remain constant, the concentration of AB must increase, i.e., the equilibrium will shift to the left hand side.

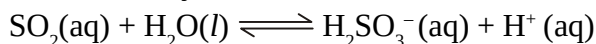
In other words, the degree of dissociation of an electrolyte (weak) is suppressed by the addition of another electrolyte (strong) containing a common ion. This is termed as common ion effect. Acetic acid is a weak electrolyte and its ionisation is suppressed in presence of a strong acid (H^+ ion as common ion) or a strong salt like sodium acetate (acetate ion as common ion). Similarly. The addition of NH_4Cl or NaOH to NH_4OH solution will suppress the dissociation of NH_4OH due to common ion either NH_4^+ or OH^- .

As a result of common ion effect, the concentration of the ion not in common in two electrolytes, is decreased. The use of this phenomenon is made in qualitative analysis to adjust concentration of S^{2-} ions in second group and OH^- ion concentration in third group of analysis.

- 172 The amount of $(NH_4)_2SO_4$ having degree of dissociation 75% which should be dissolved in 1500 ml of 1 M NH_4OH to decrease its degree of dissociation by 200 times, is -
 (A) 112.1 gm (B) 224.2 gm (C) 56.0 gm (D) 65.4 gm
- 173 500 ml of 0.2 M H_2SO_4 is mixed with 1500 ml of 0.1 M HCOOH solution. If K_a for formic acid is 1.8×10^{-4} , then pH of the final solution is -
 (A) 0.02 (B) 0.1 (C) 1 (D) 2
- 174 60 gm of CH_3COOH , 80 gm of NaOH, 58 gm of $Mg(OH)_2$ and 127.75 gm of HCl are mixed in a container and the volume of the final solution is made to 10 litres. Calculate the pH of the final solution.
 (A) 11.26 (B) 9.26 (C) 2.74 (D) 4.74

Passage-59

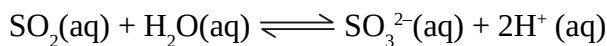
The pH value for pure water is 7.0 at 27°C, whereas natural rain water is slightly acidic. This is mainly due to presence of CO_2 in atmosphere. In many cases, however rain water is more acidic due to presence of SO_3 and NO_2 gas obtained by oxidation of SO_2 & NO gas. In some cases acid rain due to SO_3 and NO_2 consist of pH of 4.5 to a value as low as 1.7 at 27°C the acidity constants are



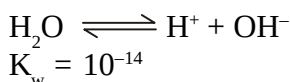
$$K_{a_1} = \frac{[\text{H}^+][\text{HSO}_3^-]}{[\text{SO}_2]} = 10^{-2} \text{ M}$$



$$K_{a_2} = \frac{[\text{SO}_3^{2-}][\text{H}^+]}{[\text{HSO}_3^-]} = 10^{-7} \text{ M}$$



$$K_a = K_{a_1} K_{a_2} = 10^{-9} \text{ M}$$



$$K_w = 10^{-14}$$

175. The Ph of 0.01 M aqueous solution of Na_2SO_3 will be
 (A) 8.5 (B) 9 (C) 9.25 (D) 9.5
176. If rain water consists of 2.463 litre SO_3 gas dissolved in 1 litre water at partial pressure of 2 atm, then pH of solution is (R = 0.0821 litre atm/K mole) ($\log 2 = 0.3$)
 (A) 0.7 (B) 1.4 (C) 0.6 (D) 0.4
177. In acid rain in Assam it was found that 1 litre rain water consist of 2.463 litre equimolar mixture of SO_3 and NO_2 gas dissolved at a partial pressure of 0.1 atm. Find out the pH of the solution if HNO_2 produced is 60% dissociated under given condition at 27°C . (R = 0.0821 litre atm/mole K) $\log 1.8 = 0.25$
 (A) 2 (B) 2.75 (C) 2.25 (D) 1.75

Passage-60

Buffer Solutions : In general, a buffer solution is one which is resistant to change of pH upon the addition of a small amount of acid or alkali. Such solutions usually consist of a mixture of a weak acid and salt of its conjugate base, e.g. HAc and NaAc (called acidic buffers), or of a weak base and salt of its conjugate acid, e.g. NH_4OH and NH_4Cl (called basic buffers). A salt of a weak acid and a weak base, e.g. NH_4Ac , also has a buffer action.

The pH of an acidic buffer and basic buffer is expressed by Henderson-Hasselbalch equations

$$\text{pH} = \text{pK}_a + \log \frac{[\text{conjugate base}]}{[\text{weak acid}]} \quad \text{pOH} = \text{pK}_b + \log \frac{[\text{conjugate acid}]}{[\text{weak base}]}$$

But limitations of the above equations is that it works only for buffer solutions that fit the following criteria

- * The ratio, $[\text{conjugate base}] / [\text{weak acid}]$ has a value between 0.1 and 10.
- * Both $[\text{conjugate base}]$ and $[\text{weak acid}]$ exceed K_a by a factor of 100 or more.

The resistance to the change in pH, on the addition of an acid or alkali is called 'Buffer action'. This

buffer action is measured by 'Buffer capacity' (β). It is expressed as $\beta = \frac{db}{dpH}$

where db is no. of moles of base added to one litre buffer solution and dpH is change in pH. The buffer capacity (β) is maximum when the acid and the salt are present in equal concentrations.

178. The acetate-acetic acid buffer is suitable in the pH range (assume K_b of $\text{CH}_3\text{COO}^- = 10^{-9}$)
 (A) 8 – 10 (B) 4 – 6 (C) 5 – 7 (D) 5.5 – 7.5
179. A solution of pH = 6 will be maintained by acidic buffer. Below are the few options given to a chemist. Which option will the chemist choose ?
 (A) HA (0.1 M), NaA (1M) K_b of $\text{A}^- = 10^{-9}$ (B) HB (0.01 M), NaB (1 M) K_b of $\text{B}^- = 10^{-10}$
 (C) HC (1 M), NaX (0.01 M) K_b of $\text{X}^- = 10^{-6}$ (D) Any one of the above
180. pH of a mixture of 1 M benzoic acid ($\text{pK}_a = 4.20$) and 1 M $\text{C}_6\text{H}_5\text{COONa}$ is 4.5 what is the volume of benzoic acid required to prepare a 300 ml buffer [$\log 2 = 0.3$] ?
 (A) 200 ml (B) 150 ml (C) 100 ml (D) 50 ml

Passage-61

In a mixture of strong acid and a strong base, concentration of H^+ ions or OH^- ions can be calculated as

$[\text{H}^+]$ or $[\text{OH}^-] = \text{meq of S.A} - \text{meq of S.B} / \text{total volume in ml of the mixture or volume given}$

At equivalence point of titration of a strong acid with strong base pH=7 at 25°C. Salt formed does not undergo salt hydrolysis.

In case of weak acid with strong base, before the equivalence point, pH is calculated using the concept of acid buffer. At the equivalence point pH is calculated taking into consideration salt hydrolysis. $\text{pH} = \frac{1}{2}(\text{pK}_w + \text{pK}_a + \log C)$

Beyond the equivalence point, there is excess of base, which mainly determines the pH. In case of weak base with strong acid, before the equivalence point buffer concept is used, at the equivalence point salt hydrolysis is taken into consideration.

$\text{pH} = \frac{1}{2}(\text{pK}_w - \text{pK}_b - \log C)$

In case of titration of weak acid and weak base at equivalence point pH is calculated on the basis of salt hydrolysis $\text{pH} = \frac{1}{2}(\text{pK}_w + \text{pK}_a - \text{pK}_b)$

181. 100ml of 0.1 M mono acid weak base, $K_b = 2 \times 10^{-5}$ is titrated with 0.1M HCl. pH of the reaction mixture at the titre value of 50ml and 100 ml are respectively.
 (A) 9.3, 9.3 (B) 9.3, 6.6 (C) 9.3, 5.3 (D) 4.7, 6.6
182. when 50ml of 0.2M HA, $K_a = 1.0 \times 10^{-5}$ is mixed with 50ml of 0.2M KOH at 25°C pH of the resulting mixture would be
 (A) 9.0 (B) 9.15 (C) 8.50 (D) 8.2
183. 100ml of 0.1M HCl is titrated with 0.1M NaOH. The pH of the reaction mixture after the addition of 50ml, 100ml, 150ml are respectively
 (A) 1.3, 7.0, 9.23 (B) 1.48, 7.0, 12.3 (C) 1.7, 7.0, 12.3 (D) 7.0, 1.3, 12.3

Passage-62

K_{sp} of $\text{CdS} = 8 \times 10^{-27}$, K_{sp} of $\text{ZnS} = 1 \times 10^{-21}$, K_a of $\text{H}_2\text{S} = 1 \times 10^{-7}$

184. Which metal sulphide is precipitated first when H_2S is added
 (A) ZnS (B) CdS
 (C) Both at same time (D) Both do not form ppt.
185. What $[\text{H}^+]$ must be maintained in a saturated H_2S (0.1M) to precipitate CdS but not ZnS, if
 $[\text{Cd}^{+2}] = [\text{Zn}^{+2}] = 0.1\text{M}$ initially
 (A) $> 0.1\text{M}$ (B) $> 0.01\text{M}$ (C) $> 0.2\text{M}$ (D) $> 0.02\text{M}$

186. When ZnS starts precipitating. What is the concentration of Cd^{+2} is left

- (A) 8×10^{-7} (B) 0.1 (C) 4×10^{-10} (D) 2×10^{-9}

Passage-63

The solution whose p^H or p^{OH} is not affected by the addition of small amount of acid or base is called buffer. Buffer is a mixture of weak acid and its salt with strong base or weak base and its salt with strong acid

$$\text{Acidic buffer } p^H = p^{K_a} + \log \left[\frac{\text{salt}}{\text{acid}} \right], \text{ Basic buffer } p^{OH} = p^{K_b} + \log \left[\frac{\text{salt}}{\text{base}} \right]$$

187. What is the p^H of the solution when 0.2M of HCl is added to one litre 1M each of CH_3COOH and acetate ion

- (A) 5.57 (B) 4.57 (C) 3.57 (D) 7.57

188. A solution of a mono basic acid was titrated with $NaOH$ solution and the end point came when 36.12ml of 0.1N $NaOH$ was added. Further 18.06ml of 0.1N HCl was added and the p^H was then found to be 4.92. Find the K_a of the acid

- (A) 1.2×10^{-5} (B) 1.2×10^{-6} (C) 2.4×10^{-5} (D) 2.4×10^{-6}

189. A certain buffer solution contains equal concentrations of X^- and HX . The K_b for X^- is 10^{-10} .

The p^H of the buffer is

- (A) 4 (B) 10 (C) 7 (D) 14

QUALITATIVE ANALYSIS

Passage-64

Pyrolusite on heating with KOH in the presence of air gives a dark green compound (A). The solution of (A) on treatment with H_2SO_4 gives a purple coloured compound (B), which gives following reactions:

- (i) KI on reaction with alkaline solution of (B) changes into a compound (C)
- (ii) The colour of compound (B) disappears on treatment with the acidic solution of $FeSO_4$.
- (iii) With conc. H_2SO_4 compound (B) gives (D) which can decompose to yield (E) and oxygen.

190. The compound (C) is

- (A) I_2 (B) I_2O_5 (C) KIO_3 (D) KIO_4

191. The compound (E) is:

- (A) MnO (B) MnO_2 (C) Mn_2O_3 (D) Mn_3O_4

192. Oxidation state of manganese of the compound (A) is

- (A) +2 (B) +4 (C) +7 (D) +6

Passage-65

A salt (A) on reaction with NaOH gives a gas (B), which gives black coloured water insoluble compound with Hg_2Cl_2 . After completely expelling the gas (B) from the solution, zinc dust is added and boiled. Appearance of gas (B) again observed. The salt (A) on heating gives a gaseous product (C) which supports combustion.

193. The gas (C) reacts with $NaNH_2$ to obtain

- (A) Na_2CO_3 (B) NH_2-NH_2 (C) NaN_3 (D) Na_2CN_2

194. The gas (B) reacts with excess of Cl_2 . The product obtained is

- (A) N_2 (B) NCl_3 (C) $COCl_2$ (D) $NOCl$

195. The gas (B) reacts with corrosive sublimate to obtain

- (A) $HgNH_2Cl$ (B) $HgNH_2Cl-Hg$ (C) Hg_2Cl_2 (D) K_2HgI_4

Passage-66

A green coloured compound (A) gave the following reactions :

- (i) (A) dissolves in water to give a green solution. The solution on reaction with $AgNO_3$ gives a white ppt. (B) which dissolves in NH_4OH solution and reappears on addition of dil HNO_3 . It on heating with $K_2Cr_2O_7$ and conc. H_2SO_4 produced a red gas which dissolves in $NaOH$ to give a yellow solution (C). Addition of lead acetate solution to (C) gives a yellow ppt. which is used as a paint
- (ii) The hydroxide of cation of (A) in borax bead test gives brown colour in oxidising flame and grey colour in reducing flame.
- (iii) Aqueous solution of (A) gives a black ppt. on passing H_2S gas. The black ppt. dissolves in aquaregia and gives back (A).
- (iv) (A) on boiling with $NaHCO_3$ and Br_2 water gives a black ppt. (D).
- (v) (A) on treatment with KCN gives a light green ppt. (E) which dissolves in excess of KCN to give (F). (F) on heating with alkaline bromine water gives the same black ppt. as (D).

196. The hybridization & magnetic behaviour of compound F' is

- (A) dsp^2 , dia (B) sp^3 , para (C) dsp^3 , para (D) sp^3 , dia

197.. The cpd A, D & E respectively are

- (A) $COCl_2, C_2O_3, CO(CN)_2$ (B) $NiCl_2, CO_2 O_3 CO(CN)_2$
(C) $NiCl_2, Ni_2O_3, Ni(CN)_2$ (D) $Ni(NO_3)_2 NiCl_2 Ni(CN)_2$

198. The cpd B & C respectively are & the Maximum OX state shown by the formation metal in cpd 'C' is

- (A) $AgCl, PbCrO_4$ & +6 (B) $AgBr, NaCrO$ & +4
(C) $MnCl_2, NiO_4$ & +7 (D) $NiCl_2, K_2(Ni(CN)_4)$ + 2

Passage-67

Some observations related to an unknown inorganic substance **A** are presented below.

- 1 A is a yellowish – white deliquescent solid and it sublimes on heating. It has a molecular weight of 267.
- 2 A reacts violently with water, forming solution B.
- 3 When a solution of NH_4Cl and NH_4OH is added to solution B, a white gelatinous precipitate is obtained
- 4 A sample of B also gives a curdy white precipitate C on addition of dilute nitric acid and silver nitrate solution. This white precipitate C readily dissolves when dilute NH_4OH is added, though a gelatinous white precipitate D is formed in its place with excess NH_4OH .
- 5 Precipitate D is filtered off and is dissolved in excess NaOH to give a clear solution E.
- 6 When CO_2 is passed through solution E, compound D is reprecipitated.
- 7 Substance A unchanged in dry ether, when this solution is reacted with excess LiH, it transforms to G.

199. Compound **A** is likely to be

- (A) Dimer of $FeCl_3$ (B) Dimer of $AlCl_3$ (C) Dimer of $SnCl_2$ (D) Dimer of $CrCl_3$

200. The curdy white ppt (C) is

- (A) $Al(OH)_3$ (B) $Sn(OH)_2$ (C) $Cr(OH)_3$ (D) $AgCl$

201. The compound (G) reacts with boron trichloride to form

- (A) acidic oxide of boron (B) compound containing 3c-2e bond
(C) boron nitride (D) lithium borohydride

SOLUTION

Passage-68

At any fixed temperature, the vapour phase is always richer in more volatile component as compared to the solution phase. In other words, mole fraction of more volatile component is always greater in vapour phase than in solution phase. We can also say that vapour phase is relatively richer in the component whose addition to liquid mixture results in an increase in total vapour pressure.

202. If 2 moles of A and 3 moles of B are mixed to form an ideal solution, vapour pressures of A and B are 120 and 180 mm of Hg respectively. The total vapour pressure of solution will be :
 (A) 48 mm Hg (B) 108 mm Hg (C) 156 mm Hg (D) 15.6 mm Hg
203. From the statement in question number 202, the composition of A and B in the vapour phase when the first traces of vapours are formed is :
 (A) A = 0.407 ; B = 0.592 (B) A = 0.8 ; B = 0.1
 (C) A = 0.109 ; B = 0.791 (D) A = 0.307 ; B = 0.693
204. From the statement in question number 202, at what pressure will the last traces of liquid disappear ?
 (A) 100 mm Hg (B) 130 mm Hg (C) 140 mm Hg (D) 150 mm Hg

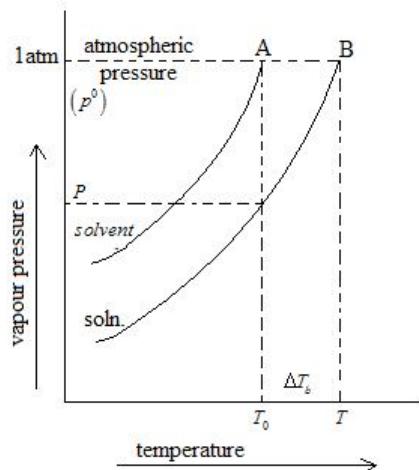
Passage-69

At any fixed temperature, the vapour phase is always richer in more volatile component as compared to the solution phase. In other words, mole fraction of more volatile component is always greater in vapour phase than in solution phase. We can also say that vapour phase is relatively richer in the component whose addition to liquid mixture results in an increase in total vapour pressure.

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 (C) A = 0.109 ; B = 0.791 (D) A = 0.307 ; B = 0.693
207. From the statement in question number 202, at what pressure will the last traces of liquid disappear ?
 (A) 100 mm Hg (B) 130 mm Hg (C) 140 mm Hg (D) 150 mm Hg

Passage-70

Figure explains elevation in boiling point when a non-volatile solute is added to a solvent



208. Elevation in boiling point of an aqueous urea solution is 0.52° (K_b of water is $0.52 \text{ K Kg mol}^{-1}$). Hence mole fraction of urea in this solution is
(A) 0.982 (B) 0.0567 (C) 0.943 (D) 0.018
209. A complex of iron and cyanide ions is 100% ionized at one molal solution. If the elevation in boiling point is 2.08°C (K_b for water $0.52 \text{ K Kg mol}^{-1}$) then the complex is
(A) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (B) $\text{Fe}(\text{CN})_2$ (C) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (D) $[\text{Fe}(\text{CN})_4]^-$
210. Ratio of $\frac{\Delta T_b}{K_b}$ of 6% AB_2 and 9% A_2B (both are non electrolytes) is one mole/Kg in both the cases. Hence atomic masses of A and B are respectively.
(A) 60, 90 (B) 40, 40 (C) 40, 10 (D) 10, 40

Passage-71

When a volatile substance A ($P_A^0 = 400 \text{ mm}$) is added to another volatile substance B ($P_B^0 = 500 \text{ mm}$), the vapour pressure of the resulting solution is 470 mm of Hg. To this solution a third volatile solvent C ($P_C^0 = 600 \text{ mm}$) is added and vapour pressure of final solution is 496 mm of Hg.

211. Mole fraction of A in the final solution
(A) 0.20 (B) 0.22 (C) 0.24 (D) 0.26
212. Mole fraction of A in the vapour phase
(A) 0.2 (B) 0.22 (C) 0.24 (D) 0.26
213. Mole fraction of C in the vapour phase
(A) 0.2 (B) 0.22 (C) 0.24 (D) 0.26

Passage-72

Dilute solutions show colligative properties. These properties depend on the number of particles of the solute but not on their nature. If solute undergoes dissociation (or) association. The colligative property differ considerably from the calculated value.

214. Which of the following solutions has highest boiling point
(A) 1% (w/v) glucose (B) 1% (w/v) fructose (C) 1% (w/v) sucrose (D) 1% (w/v) urea
215. For a solute undergoing association, the Van't Hoff's factor is
(A) $i > 1$ (B) $i < 1$ (C) $i = 0$ (D) $i = 1$
216. Solutions having same vapour pressure are called
(A) isotonic solutions (B) isopiestic solutions (C) isochoric solutions (D) isoelectric solutions.

THERMOCHEMISTRY

Passage-73

Enthalpy of Neutralisation is defined as the enthalpy change when one mole of H^+ in dilute solution combines with one mole of OH^- to give rise to undissociated water i.e.

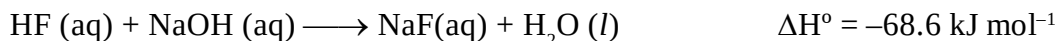


When ever one mole of strong monoprotic acid is mixed with one mole of strong base, the above neutralisation reaction takes place.

When ever a weak acid (or) base reacts with a strong base (or) acid, the release of heat is less than $57.54 \text{ kJ mol}^{-1}$. It is because of the fact that these acids (or) bases are not completely ionized in the solution. Some heat is consumed in ionising these acids and bases.

217. When 1 equivalent reacts with 0.33 equivalent of HCl, enthalpy change is -4.6 cal . When 0.66 equivalent of HCl is further added, enthalpy change would be:
 (A) -13.7 (B) -9.2 cal (C) 13.7 cal (D) 9.2 cal

218. In the following reaction, involving neutralisation of HF (a weak acid) with NaOH (a strong base).

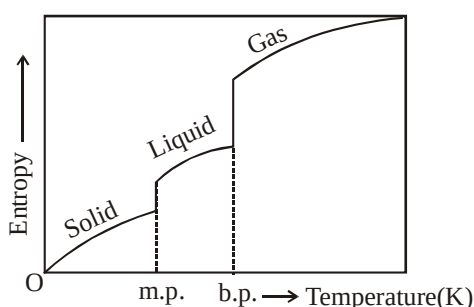


This values is much higher than the heat of neutralisation of strong acid with strong base ($= -57.3 \text{ kJmol}^{-1}$). This is due to :

- (A) in aqueous solution, HF behaves as a strong acid
 (B) there is hydration of F^- ion in aqueous solution, and process is exothermic
 (C) both (a) and (b) (D) none of the above
219. Heat of dissociation of CH_3COOH is $0.005 \text{ kcal g}^{-1}$, hence enthalpy change when 1 mole of $Ca(OH)_2$ is completely neutralised by CH_3COOH is:
 (A) -27.4 kcal (B) -13.6 kcal (C) -26.8 kcal (D) -27.1 kcal

Passage-74

Entropy is associated with the orientation and distribution of molecules in space. Disordered crystals have higher entropy than ordered crystals and diffused gases have higher entropy than compressed gases.



Entropy is also associated with molecular motion. As the temperature of a substance increases, random molecular motion increases hence entropy increases. Figure gives variation of entropy with temperature. At absolute zero (-273°C) every substance is in solid state whose particle are rigidly fixed in a crystalline structure. If there is no residual orientational disorder, like that in CO, entropy of the substance at 0 K will be zero. Third law of thermodynamics states that. **“At the absolute zero of temperature the entropy of every substance may become zero and does become zero in case of perfectly crystalline structure.”**

$$\lim_{T \rightarrow 0} s = 0$$

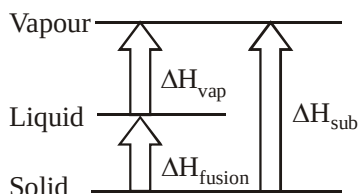
$$T \rightarrow 0$$

In case of CO and NO molecules in solid state, there is randomness even at 0 K due to their

dipole moment hence entropy in such cases is not zero even at 0 K.

As the temperature is raised, the molecules begin to vibrate. The number of ways in which the vibrational energy can be distributed increases with increase in temperature and the entropy of the solid increases steadily as the temperature increases. At the melting point (m.p.) of a solid, there is a discontinuous jump in entropy because there are many more ways of arranging the molecules in the liquid than in the solid. An even greater jump in entropy is observed at the boiling point (b.p.) because molecules in the gas are free to occupy a more larger volume and randomness increases.

220. From the figure representing enthalpy change of various transition as indicated, a substance has maximum enthalpy when present in:



- (A) solid state (B) liquid state
(C) gaseous state (D) equal in all states
221. For the following reaction, $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
If at 0 K, ΔS of this reaction is $x \text{ cal mol}^{-1}$ then entropy of $\text{CO}_2(\text{g})$ at 0 K is :
(A) $-x \text{ cal}$ (B) $-2x \text{ cal}$ (C) $+x \text{ cal}$ (D) 0 cal
222. Which has the highest entropy per mol of the substance?
(A) H_2 at 25°C at 1 atm (B) H_2 at STP
(C) H_2 at 100 K at 1 atm (D) H_2 at 0 K at 1 atm

Passage-75

The first law of thermodynamics is given as $q = \Delta E - W$; where 'q' is heat given to the system and ΔE represents increase in internal energy and - 'W' is the work done by the system. Various processes such as isothermal, adiabatic, iso-choric and iso-baric process in terms of first law of thermodynamics leads to important results. Work done by the system in isothermal reversible process is given by; W_{rev}

$= -nRT \ln \left(\frac{V_2}{V_1} \right)$. Also in case of Adiabatic reversible process the work done by the system is given

by $W_{\text{rev}} = \frac{nR}{(\gamma - 1)} (T_i - T_f)$. Thus if $T_f > T_i$, then since $1 - \gamma = \text{negative}$ (since $\gamma > 1$) the work done is positive i.e work is done on the gas. Also if $T_f < T_i$, the work done is negative i.e work is done by the gas. Adiabatic process obey $PV^\gamma = \text{constant}$, whereas in an isothermal process, $PV = \text{constant}$.

223. 20 grams of Neon (At. Wt = 20) initially at a pressure of 506.625 kPa and temperature of 475K expanded adiabatically to a pressure of 50.6625 kPa. Which of the following is/are correct, when the expansion is a free expansion?
i. $W = 0$ ii. $q = 0$ iii. $dE = 0$
iv. Temperature of the gas remains unchanged during the expansion.
(A) i, ii and iii (B) ii, iii and iv (C) i, ii, iii and iv (D) iv only
224. Calculate the work done when 5 moles of an ideal gas at 300K are expanded isothermally from an initial pressure of $4 \times 10^5 \text{ Pa}$ to a final pressure of $1 \times 10^5 \text{ Pa}$ against a constant external pressure of 10^5 Pa . ($5 \times 8.314 \times 300 = 12470$)
(A) -9.35 kJ (B) -16.9 kJ (C) Zero (D) $+16.9 \text{ kJ}$

225. In a particular experiment, a gas undergoes adiabatic expansion satisfying the equation $VT^3 = \text{constant}$. The ratio of specific heats " γ " : is

(A) 4 (B) 3 (C) $\frac{5}{3}$ (D) $\frac{4}{3}$

Passage-76

Entropy is the degree of disorder or randomness of the molecule of a system.

Characteristic property of a spontaneous process is always accompanied by an increase in the randomness of the molecules of the system.

The entropy change depends only on the initial and final states and not on the path taken by the system.

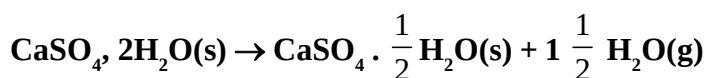
226. Entropy is _____ function
(A) state (B) path (C) state and path (D) None
227. The entropy and entropy change for pure crystalline solid substance at $0^\circ K$ is
(A) increases (B) decreases (C) zero (D) Maximum
228. The increase in entropy of the system is _____ in any reversible cyclic process
(A) Maximum (B) Minimum (C) zero (D) 100

Passage-77

On July 1, 2000, the combined tunnel and bridge connecting Denmark and Sweden was officially opened. It consists of a tunnel from Copenhagen to an artificial island and a bridge from the Island to Malmo in Sweden. The major construction materials employed are concrete and steel. This problem deals with chemical reactions relating to production and degradation of such materials.

Concrete is produced from a mixture of cement, water, sand and small stones. Cement consists primarily of calcium silicates and calcium aluminates formed by heating and grinding of clay and limestone. In the later steps of cement production, a small amount of gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, is added to improve subsequent hardening of the concrete. The use of elevated temperatures during the final production may lead to formation of unwanted semihydrate, $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$.

Consider the following reaction :



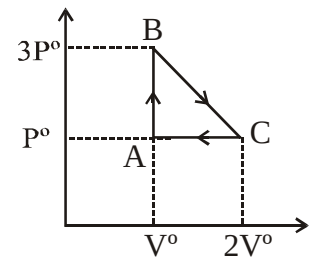
	Compound	H.(kJ mol ⁻¹)	S.(JK ⁻¹ mol ⁻¹)
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O(s)}$	2021.0	194.0	
$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O(s)}$	1575.0	130.5	
$\text{H}_2\text{O(g)}$	241.8	188.6	

229. ΔH (in kJ) for transformation of 1 kg of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O(s)}$ to $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O(s)}$ is equal to
(A) 484 kJ (B) 450 kJ (C) -484 kJ (D) -450 kJ
230. On the basis of value of ΔH
(A) If the thermodynamic state of products do not change, then reaction is endothermic in nature.
(B) If the thermodynamic state of products do not change, then reaction is exothermic in nature.
(C) If the thermodynamic state of products do not change, then there is no heat exchange between system and surrounding.
(D) none of these

231. ΔS (in J) for the Transformation of 1 mole of Gypsum to 1 mole of plaster of paris in above process is
 (A) 219.4 B) 319.4 (C) 0.219 (D) 3.19

Passage-78

One mole of ideal monoatomic gas is carried through the reversible cyclic process as shown in figure.



232. Work done by the gas

- (A) $-\frac{1}{2} P_0 V_0$ (B) $+\frac{1}{2} P_0 V_0$
 (C) $-P_0 V_0$ (D) $+P_0 V_0$

233. The ratio of heat exchanged by the gas in a path CA to AB.

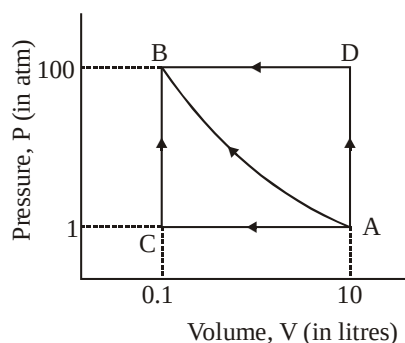
- (A) $\frac{3}{5}$ (B) $-\frac{15}{2}$ (C) $-\frac{5}{6}$ (D) $\frac{10}{3}$

234. Net heat absorbed by the gas in the path BC.

- (A) $-\frac{1}{2} P_0 V_0$ (B) $+\frac{1}{2} P_0 V_0$ (C) $-P_0 V_0$ (D) $+P_0 V_0$

Passage-79

Assume perfect gas behaviour in the following isotherm AB. Take $C_v = \frac{3}{2} R$, temperature of the isotherm = 313 K



235. Number of moles of gas in this system at the start is:
 (A) 0.39 mol (B) 0.17 mol (C) 0.78 mol (D) 0.34 mol
236. Work done on the gas along the isotherm AB is:
 (A) -4.78 kJ (B) -8 kJ (C) +4.78 kJ (D) +71 kJ
237. ΔE (Change in internal energy) for path ACB or ADB is:
 (A) 19.7 kJ (B) 0.00 kJ (C) 51.2 kJ (D) 3.0 kJ

Passage-80

Enthalpy of reaction : It is used for the heat change accompanying any reaction. depending upon the nature of the reaction the enthalpy of the reaction is named accordingly

For example : In the reaction $\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ $\Delta H = -890.4 \text{ kJ/mole}$

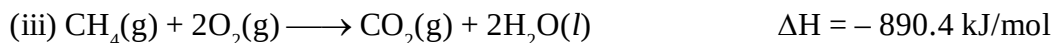
This reaction shown that 890.4 kJ of heat is produced when methane is completely burnt it is known as heat of combustion .

for example: In the reaction



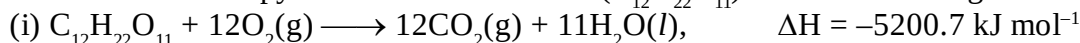
when $\text{CO}_2\text{(g)}$ is formed from its elements i.e. from C(s) and $\text{O}_2\text{(g)}$ 393.5 kJ of heat is produced and the process is heat of formation.

238. The enthalpy of formation of methane from the following data



(A) -74.9 kJ/mole (B) 54.6 kJ/mole (C) 99.7 kJ/mole (D) -89.6 kJ/mole

239. Calculate the enthalpy of formation of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) from the following data:



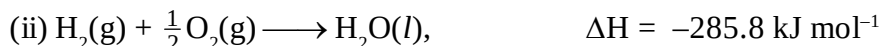
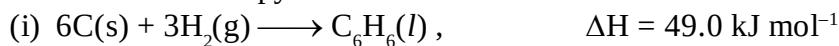
(A) -10.3 kJ/mole

(B) +863.4 kJ/mole

(C) -2677.1 kJ/mole

(D) -4321.3 kJ/mole

240. Calculate the enthalpy of combustion of benzene from the following data:



(A) +463.6 kJ/mole

(B) +32.4 kJ/mole

(C) -1432.3 kJ/mole

(D) -3242.2 kJ/mole

CHEMICAL KINETICS

Passage-81

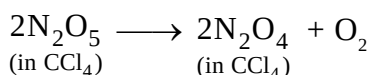
A chemical reaction occurs when the existing bonds of the reactants are broken down and new bonds come into existence in the molecules of products. The basic requirement for a reaction to occur is that there should be physical contact between reacting species, i.e., they must collide with one another. Effective collisions which bring chemical change are few in spite of large number of actual collisions. For a collision to be effective, the following two conditions must be satisfied:

- (i) Reacting species must possess adequate energy to overcome the energy barrier.
- (ii) Reacting molecules must be properly oriented at the time of collision.

The minimum amount of energy which the colliding particles must possess as to make the chemical reaction to occur is called threshold energy. The excess energy (over and above the average energy of the reacting species) required by the reacting species to undergo chemical reaction is called activation energy (E_a).

241. The rate of a reaction triples when temperature changes from 20°C to 50°C. Calculate energy of activation for the reaction ($R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$) ($\log 3 = 0.48$)
(A) 6.97 kcal/mole (B) 3.21 kcal/mole (C) 8.14 kcal/mole (D) 7.25 kcal/mole

242. The energy of activation and specific rate for a first order reaction at 27°C



are 23 kcal/mole and $4.81 \times 10^{-5} \text{ sec}^{-1}$ respectively. Determine the temperature at which the half life of the reaction is 2 hours ($\log 2 = 0.3010$)

- (A) 305.5 K (B) 350 K (C) 325 K (D) 310 K
243. Two reactions proceed at 25°C at the same rate. The temperature coefficient of the rate of the first reaction is 2 and that of the second is 2.5. Find the ratio of the rates of these reactions at 95°C.
(A) 5.362 (B) 3.184 (C) 1.324 (D) 4.768

Passage-82

Order of reaction is an experimentally determined quantity. It may be zero, positive, negative and fractional. The kinetic equation of nth order reaction is:

$$k \times t = \frac{1}{(n-1)} \left[\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}} \right]$$

Half life of nth order reaction depends on initial concentration according to the following relation:

$$t_{1/2} \propto \frac{1}{a^{n-1}}$$

Unit of the rate constant varies with the order but general relation for unit of nth order reaction is:

$$\text{Unit of } k = \left[\frac{1}{\text{conc.}} \right]^{n-1} \times \text{time}^{-1}$$

The differential rate law for nth order reaction may be given as:

$$\frac{dx}{dt} = k[A]^n$$

Where, A denotes the reactant.

Answer the following questions:

244. The rate constant for zero order reaction is (where C_0 & C_t are concentrations of reactants at respective time):

(A) $k = \frac{C_0}{2t}$ (B) $k = \frac{C_0 - C_t}{t}$ (C) $k = \ln \frac{C_0 - C_t}{2t}$ (D) $k = \frac{C_0}{C_t}$

245. The half life for a zero order reaction equals to:

- (A) $\frac{1}{2} \frac{k}{a^2}$ (B) $\frac{a^2}{2k}$ (C) $\frac{2k}{a}$ (D) $\frac{a}{2k}$

246. For a reaction: $I^- + OCl^- \rightarrow IO^- + Cl^-$
in an aqueous medium, the rate of the reaction is given by

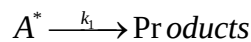
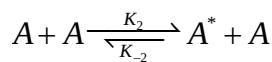
$$\frac{d[IO^-]}{dt} = k \frac{[I^-][OCl^-]}{[OH^-]}$$

The overall order of the reaction is:

- (A) -1 (B) 1 (C) zero (D) 2

Passage-83

A chemical reaction is proceeding as



To study the kinetics, steady state approximation is used.

247. The rate of disappearance of A is

- (A) $K_2[A]^2 - K_2[A]^2$ (B) $K_2[A]^2 - K_2[A]^2$
(C) $K_2[A]^2 - K_2[A][A^*]$ (D) $K_2[A][A^*] - K_2[A]^2$

248. The conditions under which this reaction is a Pseudo-first-order reaction

- (A) $\frac{K_2}{K_1}[A] \gg 1$ (B) $\frac{K_2}{K_1}[A] \ll 1$ (C) $\frac{K_2}{K_1}[A] \gg 1$ (D) $\frac{K_2}{K_1}[A] \ll 1$

249. The conditions under which the reaction is Pseudo-Second order reaction

- (A) $\frac{K_2}{K_1}[A] \ll 1$ (B) $\frac{K_2}{K_1}[A] \gg 1$ (C) $\frac{K_2}{K_1}[A] \ll 1$ (D) $\frac{K_2}{K_1}[A] \gg 1$

Passage-84

In first order reaction, the rate is determined by the change of one concentration term only. The unit of rate constant is time^{-1} . The half-life period is independent of concentration. Acid hydrolysis of disaccharides, all radioactive decays are few examples of first order reactions.

250. Inversion of sugar follows first order rate equation which can be followed by noting the change in rotation of the plane of polarized light is the polarimeter. If r_∞ , r_t and r_0 are the rotations at $t = \infty$, $t = t$ and $t = 0$, then first order reaction can be written as

- (A) $K = \frac{1}{t} \log e \frac{r_\infty - r_t}{r_0 - r_\infty}$ (B) $K = \frac{1}{t} / n \frac{r_t - r_\infty}{r_0 - r_\infty}$
(C) $K = \frac{1}{t} / n \frac{r_\infty - r_0}{r_\infty - r_t}$ (D) $K = \frac{1}{t} / n \frac{r_\infty - r_t}{r_\infty - r_0}$

251. For a first order reaction, the graph between $\log\left(\frac{a}{a-x}\right)$ versus time is a straight line. The slope of the line equals to

- (A) $-\frac{k}{2.303}$ (B) $-k \times 2.303$ (C) $+\frac{k}{2.303}$ (D) $2.303 k$

252. The half-time of first order decomposition of nitramide is 2.1 hour at 15°C . If 6.2 g of NH_2NO_2 is allowed to decompose, the time taken to decompose 99% would be :

- (A) 13.957 h (B) 6.978 h (C) 4.606 h (D) 27.914 h

Passage-85

A compound $X_{2(g)}$ decomposed to its manner $X_{(g)}$ at high temperature. The decomposition follows first order kinetics. When certain amount of dimmer is rapidly introduced into a flask at 127°C , after 693 sec the pressure in the vessel observed was 30.0 mm and at a very long time it was observed 40.0 mm. When the same experiment is carried out at 227°C , the pressure after 220 sec observed was 75.0 mm and after a very long time it was observed 90.0 mm [$\log 2 = 0.3010$ and $\log 3 = 0.4771$]

253. The rate constant of the reaction at 127°C is

- (A) $1 \times 10^{-3} \text{ min}^{-1}$ (B) $2 \times 10^{-3} \text{ min}^{-1}$ (C) $5 \times 10^{-3} \text{ sec}^{-1}$ (D) $1 \times 10^{-3} \text{ sec}^{-1}$

254. The partial pressure of $X_2(g)$ at 227°C , after 220sec was

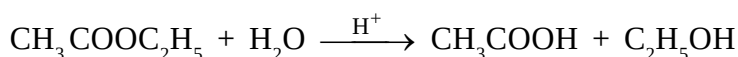
- (A) 15 mm (B) 45 mm (C) 37.5 mm (D) 75 mm

255. Activation energy of the reaction is

- (A) 52.9 KCal (B) 5.29 KCal (C) 6.44 KCal (D) 64.4

Passage-86

Hydrolysis of esters :



In this reaction acetic acid is one of the products, the amount of which can be found by titration against NaOH solution. But being an acid-catalysed reaction, the acid present originally as catalyst, also react with NaOH solution.

Hence a little careful thought reveals that for the same volume of reaction mixture withdrawn at different times.

The rate constant 'k' can be calculated as, $k = \frac{2.303}{t} \log \frac{V_\infty - V_0}{V_\infty - V_t}$

- $V_\infty \Rightarrow$ Volume of base used for acid base titration at $t = \infty$.
 $V_0 \Rightarrow$ Volume of base used for acid base titration at $t = 0$.
 $V_t \Rightarrow$ Volume of base used for acid base titration at $t = t$

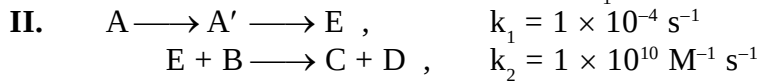
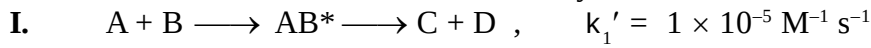
256. The acid catalysed hydrolysis of an organic compound (A) at 30°C has half life of 100 minutes when carried out in a buffer solution of pH = 5 and 10 minute when carried out at pH = 4. Both the times the half life are independent of the initial concentration of A. If the rate of reaction is given by rate = $k[A]^x[H^+]^y$, what are the values of 'x' and 'y'.

- (A) $x = 1, y = 1$ (B) $x = 2, y = 1$ (C) $x = 1, y = 2$ (D) $x = 1, y = 0$

257. If $2V_t = V_\infty + V_0$ happen at $t = 40$ minutes then % hydrolysis of the ester is:
 (A) 40 % (B) 50 % (C) 75 % (D) 90 %
258. For the same reaction between V_∞ , V_0 and V_t after 266.4 minutes will be :
 (A) $100 V_t = 99 V_\infty + V_0$ (B) $100 V_\infty = 99 V_t + V_0$
 (C) $100 V_0 = 99 V_\infty - V_t$ (D) $100 V_t = 99 V_\infty - V_0$

Passage-87

For the overall reaction between A and B to yield C and D, two mechanisms are proposed ;



[species with * are short lived]

259. Rate law for mechanism I when concentration of each is 0.1 M is :
 (A) $1 \times 10^{-7} \text{ M s}^{-1}$ (B) $1 \times 10^{-6} \text{ M s}^{-1}$
 (C) $1 \times 10^{-5} \text{ M s}^{-1}$ (D) $1 \times 10^{-4} \text{ M s}^{-1}$
260. Rate law for mechanism II when concentration of each is 0.01 M is :
 (A) $1 \times 10^{-4} \text{ M s}^{-1}$ (B) $1 \times 10^{-8} \text{ M s}^{-1}$
 (C) $1 \times 10^{-6} \text{ M s}^{-1}$ (D) $1 \times 10^{-10} \text{ M s}^{-1}$
261. At what concentration of B, rates of two mechanism are equal .
 (A) 1 M (B) 5 M (C) 7 M (D) 10 M

Passage-88

$t_{1/2}$ for n^{th} order reaction can be calculated as
 $nA \rightarrow \text{products}$

$$\text{So, rate} = \frac{-d[A]}{dt} = K_n[A]^n$$

$$\text{or} \quad - \int \frac{d[A]}{[A]^n} = K_n \int dt$$

$$t_{1/2} = \frac{2^{n-1} - 1}{K_n(n-1)a_0^{n-1}}$$

where a_0 = initial concentration of A

$$\text{So, } t_{1/2} \propto \frac{1}{a_0^{n-1}}$$

262. For a reaction of n^{th} order, the ratio of $\frac{t_{1/2}}{t_{3/4}}$ is a function of
 (A) a_0 (B) K_n (C) n (D) t
263. The following data was obtained for a chemical reaction

$a_0(\text{m/l})$	0.5	1.1	2.48
$t_{1/2}(\text{second})$	4280	885	174

 the order of reaction is:
 (A) 1 (B) 2 (C) 3 (D) 4
264. Which of these statement is true?
 (A) $t_{1/2}$ increases with increase in temperature
 (B) $t_{1/2}$ decreases with increase in temperature
 (C) $t_{1/2}$ is independent of temperature
 (D) $t_{1/2}$ is time taken when half the actual time is taken for the chemical reaction to complete

Passage-89

Concentration units :

A solution is defined as a homogenous mixture of two (or) more chemically non-reacting substances; the relative amounts of which can be varied upto a certain limit. The solution contains two components solute and solvent. the concentration can be expressed in a number of ways

(i) strength (ii) molarity (iii) molality (iv) mole fraction

265. In a reaction vessel 0.184 g of NaOH is required for completing the reaction. How many millilitres of 0.15 M NaOH solution should be added for this requirement:
(A) 30.7 ml (B) 35.5 ml (C) 61.4 ml (D) 17.25 ml
266. 250 ml of a sodium carbonate solution contains 2.65 gm of Na_2CO_3 . If 10 ml of this solution is diluted to one litre, what is the concentration of the resultant solution.
(A) 0.1 M (B) 0.001 M (C) 0.01 M (D) 0.0001 M
267. Commercially available concentrated HCl contains 38% HCl by mass. What is the molarity of this solution of the density is 1.19 g cm^{-3} .
(A) 10.45 M (B) 5.67 M (C) 12.38 M (D) 6.19 M

Passage-90

In the single-reactant reactions, the calculations are carried out with only that amount of the reactant which has converted to the product. This is done in all the methods mentioned above. In the reactions where more than one reactant are involved, one has first to identify the limiting reactant, i.e., the reactant which is completely consumed. All calculations are to be carried out with the amount of the limiting reactant only.

268. The reaction, $2\text{C} + \text{O}_2 \longrightarrow 2\text{CO}$ is carried out taking 24 g of carbon and 96 g of O_2 . Find out how many g of the other reactant should be taken so that nothing is left at the end of the reaction?
(A) 72 g (B) 56 g (C) 96 g (D) 48 g
269. Calculate the weight of FeO produced from 2 g of VO and 5.75 g of Fe_2O_3 for the reaction
 $\text{VO} + \text{Fe}_2\text{O}_3 \longrightarrow \text{FeO} + \text{V}_2\text{O}_5$ [at. wt. of V = 51, Fe = 56]
(A) 5.17 g (B) 8.35 g (C) 4.32 g (D) 2.585 g
270. How much chlorine gas can be obtained by the reaction of 3.48 g MnO_2 and 5.6 g of HCl?
 $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$ (molecular weight of Mn = 55)
(A) 2.71 g (B) 8.62 g (C) 5.63 g (D) 4.32 g

Passage-91

Acid base reaction is an important part of mole concept. It is a non-redox reaction and also known as neutralisation reaction. Because acid base neutralises each other and forms water. But when base is ammonia, only salt is formed.

For eq. $\text{HCl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$

$\text{NH}_3 + \text{HCl} \longrightarrow \text{NH}_4\text{Cl}$

If both the compounds are basic or acidic in nature then they can exist in solution. But if one is acidic and another is basic then they will neutralise other.

271. When 5 moles of $\text{Mg}(\text{HCO}_3)_2$ reacts with 980 gm of H_2SO_4 then volume of CO_2 gas produced at STP is :
(A) 112 l (B) 224 l (C) 448 l (D) 56 l

272. When NH_4Cl reacts with excess of $\text{Mg}(\text{OH})_2$, weight of NH_3 produced is 13.6 g then mass of NH_4Cl reacted is :
 (A) 23.4 g (B) 93.6 g (C) 42.8 g (D) 11.7 gm
273. Volume of NH_3 at STP required to neutralise 73 g of HCl is :
 (A) 44.8 l (B) 22.4 l (C) 11.2 l (D) 5.6 l

Passage-92

Redox reactions are those in which oxidation and reduction take place simultaneously. Oxidising agent can gain electron whereas reducing agent can lose electron easily. The oxidation state of any element can never be in fraction. If oxidation number of any element comes out to be in fraction, it is average oxidation number of that element which is present in different oxidation states.

274. What are the oxidation states of N in NH_4NO_3 ?
 (A) +3, +2 (B) +1, -2 (C) -3, +5 (D) +2, -2
275. How many moles of electrons are needed for the reduction of each mole of Cr in the reaction?

$$\text{CrO}_5 + \text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O} + \text{O}_2$$
 (A) 4 (B) 3 (C) 5 (D) 7
276. The percentage of Pb^{2+} in Pb_3O_4 is
 (A) 66.7 % (B) 33.3 % (C) 50% (D) 25%

Passage-93

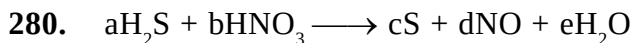
Whenever any solute is dissolved in a given solvent, a solution is formed. Solution may be homogeneous or heterogeneous. Homogeneous solution is termed as true solution and in this concentration of solution is defined as mole or mass of solute present in the given weight or volume of solution. Units which depend on mass are temperature independent while which depend on volume are temperature dependent. Answer following question after reading passage—

277. Which of the following unit is temperature dependent
 (A) molality (B) ppm (C) mole fraction (D) molarity
278. If aqueous solution of KOH contains 28% by weight of KOH hence mole fraction of KOH is
 (A) $\frac{1}{9}$ (B) $\frac{8}{9}$ (C) $\frac{10}{9}$ (D) $\frac{2}{9}$
279. Mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) present in 50 ml of its 2 molar solution is
 (A) 18000 gm (B) 180 gm (C) 18 gm (D) 1800 gm

Passage-94

Redox reactions involve simultaneous reduction-oxidation reactions. The process of oxidation involves addition of oxygen or any other electronegative element or loss of hydrogen or any other electropositive element. The reverse of this process is called reduction. Reduction also involves addition of electrons or decrease in the oxidation number of an atom or ion present in a substance. The reverse of this defines the process of oxidation. Substances which bring out oxidation of other substance are called oxidants while those which bring out the reduction of other substances are called reductants. In terms of electronic concept, reductants are electron donors while oxidants are electron acceptors. Oxidants also involve decrease in the oxidation number of one of its atoms/ions while reductants involve increase in the oxidation number of one of its atoms/ions. Oxidation numbers are always whole numbers and must be always calculated on the basis of their structures and never from their molecular formulae. Redox reactions may involve combination of atoms/molecules, decomposition of substances, displacement of metals or non-metals and disproportionation of a particular species which may be metals, non-metals

or ions. Redox reactions can be balanced both by oxidation number method as well as ion-electron method

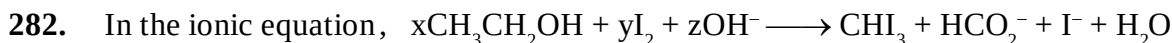


The value of $\frac{c \times d - e}{a - b}$

- (A) 1 (B) 2 (C) 4 (D) 8

281. Which among the following are not an auto redox reaction

- (A) $\text{P}_4 + \text{OH}^- \longrightarrow \text{H}_2\text{PO}_4^- + \text{PH}_3$ (B) $\text{S}_2\text{O}_3^{2-} \longrightarrow \text{SO}_4^{2-} + \text{S}$
(C) $\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2$ (D) $\text{AgCl} + \text{NH}_3 \longrightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl}$



- (A) $x = 1, y = 4, z = 6$ (B) $x = 2, y = 6, z = 4$
(C) $x = 2, y = 5, z = 10$ (D) $x = 3, y = 8, z = 12$

Passage-95

When metal carbonates are strongly heated, in general, metal oxide and CO_2 are obtained. However, carbonates such as Na_2CO_3 are thermally stable and do not undergo thermal decomposition. Bicarbonates such as NaHCO_3 upon heating results their respective carbonates alongwith CO_2 . Ag_2CO_3 on the other hand when heated leaves behind a residue of Ag alongwith release of CO_2 and O_2 .

283. 0.2 moles of Na_2CO_3 and 0.4 moles of NaHCO_3 are strongly heated volume of CO_2 released will be

- (A) 1.12 lit. (B) 4.48 lit. (C) 11.2 lit. (D) 22.4 lit.

284. 0.4 moles each of MgCO_3 and CaCO_3 and strongly heated. CO_2 released is allowed to dissolve in H_2O . Number of H^+ obtained from formed H_2CO_3 upon complete dissociation will be (where N_A is Avagadro's number)

- (A) $0.8 N_A$ (B) $1.6 N_A$ (C) $2 N_A$ (D) $0.4 N_A$

285. 13.8 g of Ag_2CO_3 is strongly heated. If released gases are used for combustion of C_2H_2 , the number of moles of C_2H_2 oxidised are

- (A) $\frac{5}{2} \times 10^{-2}$ (B) 10^{-2} (C) 2×10^{-2} (D) 5×10^{-2}

Passage-96

8.7 gm of a sample of MnO_2 (molar mass = 87) is used in a chemical reaction with HCl to release Cl_2 . Released Cl_2 is used to displace I_2 from excess of KI solution. The iodine hence released is estimated using hypo and consumes 100 ml of hypo solution.. 12.5 ml of same hypo solution is required for complete reaction with 25 ml of 0.5 N I_2 solution.

286. % purity of MnO_2 is

- (A) 4.35 (B) 43.5 (C) 50 (D) 6.25

287. Wt. of Na_2CO_3 which may be completely neutralized by the amount of HCl made to react quantitatively with pure MnO_2 in the sample is

- (A) 106 g (B) 10.6 g (C) 5.3 g (D) 53 g

288. Number of milli moles of iodine in the solution used with hypo are

- (A) 6.25 (B) 12.5 (C) 10 (D) 20

Passage-97

The E^0 (standard Reduction Potential) values are given below

	$E^0(V)$		$E^0(V)$
$Mn^{2+} + 2e^- \rightarrow Mn$	-1.18	$Cu^{2+} + 2e^- \rightarrow Cu$	+0.34
$Zn^{2+} + 2e^- \rightarrow Zn$	-0.76	$Hg_2^{2+} + 2e^- \rightarrow 2Hg$	+0.79
$Fe^{2+} + 2e^- \rightarrow Fe$	-0.44	$Ag^+ + 1e^- \rightarrow Ag$	+0.80

289. Which one of the metals does not displace hydrogen from dilute acids ?

- (A) Hg (B) Zn (C) Mn (D) Fe

290. Which one of the following metal ions can be reduced easily ?

- (A) Zn^{2+} (B) Ag^+ (C) Fe^{2+} (D) Mn^{2+}

291. Which one of the following process has ΔG^0 as negative value ?

- (A) Reduction of Cu^{2+} (B) Reduction of Mn^{2+}
(C) Reduction of Zn^{2+} (D) Reduction of Fe^{2+}

Passage-98

3.6 gm of magnesium is burnt in limited supply of oxygen. The residue was treated with 100 ml of H_2SO_4 (35% by mass, 1.26 g/ml density) and 2.463 litre of H_2 at 1 atm and $27^\circ C$ was evolved. After the reaction, H_2SO_4 was found to have a density of 1.05 g/ml. Assuming no volume change in H_2SO_4 .

292. The final % by wt. of H_2SO_4 was

- (A) 25% (B) 28% (C) 36.78% (D) 30%

293. The % by mass of magnesium converted to oxide

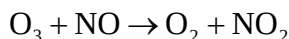
- (A) 25% (B) 30% (C) 33.33% (D) 50%

294. Volume of the used O_2 at STP was—

- (A) 1.12 litre (B) 0.56 litre (C) 2.24 litre (D) 4.89 litre

Passage-99

The depletion of ozone (O_3) in the stratosphere has been a matter of great concern among scientists in recent years. It is believed that ozone can react with nitric oxide (NO) that is discharged from the high altitude jet plane. The reaction is:



295. If 0.740 g of O_3 reacts with 0.670 g of NO, how many gram of NO_2 will be produced ?

- (A) 0.71 g (B) 0.74 g (C) 0.68 g (D) 0.81 g

296. Which compound is the limiting reactant ?

- (A) NO (B) O_3
(C) Both are in equimolar ratio (D) Both are in stoichiometric ratio

297. Number of moles of the excess reactant remaining at the end of the reaction is

- (A) 0.007 mol O_3 (B) 0.014 mol O_3 (C) 0.007 mol NO (D) 0.014 mol NO

NUCLEAR CHEMISTRY

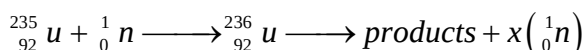
Passage-100

Nuclear fission is the process of splitting of a heavy nucleus by bombarding with thermal neutrons into two approximately equal parts, with simultaneous liberation of a huge amount of energy. If the sample of the fissionable material has a sufficiently large mass called critical mass, the neutrons produced cause additional fission reaction and a chain fission reaction ensues.

298 The missing product in the reaction : ${}_{92}^{235}\text{U} + {}_0^1\text{n} \longrightarrow {}_{57}^{146}\text{La} + ? + 3({}_0^1\text{n})$ is

- (A) ${}_{35}^{86}\text{Br}$ (B) ${}_{35}^{87}\text{Br}$ (C) ${}_{35}^{89}\text{Br}$ (D) ${}_{32}^{89}\text{Ge}$

299. The pair of elements that are produced in the following nuclear fission is



where x is either 2 or 3.

- 1) ${}_{56}^{144}\text{Ba}$ 2) ${}_{36}^{90}\text{Kr}$ 3) ${}_{54}^{144}\text{Xe}$ 4) ${}_{55}^{144}\text{Cs}$
(A) 3, 4 (B) 2, 3 (C) 1, 2 (D) 1, 4

300. Which one is correct for a nuclear fission reaction.

- (A) the n/p ratio is low for the product elements
(B) in general β - emission is observed after the product elements are formed
(C) in general α - emission is observed after the product elements are formed
(D) the product elements are not radioactive in nature.

ANSWERS

1.	(A)	2.	(A)	3.	(C)	4.	(A)	5.	(B)	6.	(A)
7.	(C)	8.	(B)	9.	(A)	10.	(A)	11.	(C)	12.	(C)
13.	(C)	14.	(A)	15.	(B)	16.	(A)	17.	(D)	18.	(C)
19.	(D)	20.	(A)	21.	(B)	22.	(C)	23.	(C)	24.	(B)
25.	(C)	26.	(C)	27.	(D)	28.	(A)	29.	(ABC)	30.	(BC)
31.	(B)	32.	(C)	33.	(B)	34.	(C)	35.	(C)	36.	(D)
37.	(C)	38.	(B)	39.	(B)	40.	(A)	41.	(B)	42.	(D)
43.	(B)	44.	(A)	45.	(D)	46.	(B)	47.	(A)	48.	(D)
49.	(C)	50.	(C)	51.	(D)	52.	(A)	53.	(D)	54.	(A)
55.	(C)	56.	(D)	57.	(C)	58.	(C)	59.	(A)	60.	(B)
61.	(C)	62.	(A)	63.	(A)	64.	(A)	65.	(D)	66.	(A)
67.	(D)	68.	(B)	69.	(C)	70.	(A)	71.	(C)	72.	(A)
73.	(A)	74.	(A)	75.	(D)	76.	(A)	77.	(B)	78.	(C)
79.	(A)	80.	(A)	81.	(D)	82.	(C)	83.	(A)	84.	(C)
85.	(D)	86.	(A)	87.	(B)	88.	(D)	89.	(A)	90.	(D)
91.	(D)	92.	(B)	93.	(C)	94.	(D)	95.	(C)	96.	(A)
97.	(A)	98.	(B)	99.	(D)	100.	(A)	101.	(B)	102.	(D)
103.	(C)	104.	()	105.	(D)	106.	(D)	107.	(D)	108.	(B)
109.	(B)	110.	(A)	111.	(C)	112.	(A)	113.	(A)	114.	(D)
115.	(D)	116.	(B)	117.	(B)	118.	(B)	119.	(A)	120.	(A)
121.	(B)	122.	(B)	123.	(A)	124.	(A)	125.	(C)	126.	(A)
127.	(D)	128.	(C)	129.	(C)	130.	(A)	131.	(B)	132.	(D)
133.	(C)	134.	(A)	135.	(A)	136.	(B)	137.	(C)	138.	(D)
139.	()	140.	(A)	141.	(C)	142.	(A)	143.	(A)	144.	(B)
145.	(A)	146.	(D)	147.	(C)	148.	(A)	149.	(B)	150.	(C)
151.	(C)	152.	(B)	153.	(A)	154.	(C)	155.	(B)	156.	(B)
157.	(A)	158.	(B)	159.	(A)	160.	(A)	161.	(A)	162.	(B)
163.	(A)	164.	(B)	165.	(B)	166.	(C)	167.	(C)	168.	(C)
169.	(C)	170.	(B)	171.	(B)	172.	(A)	173.	(C)	174.	(D)
175.	(D)	176.	(D)	177.	(D)	178.	(B)	179.	(A)	180.	(C)
181.	(C)	182.	(A)	183.	(B)	184.	(B)	185.	(A)	186.	(A)
187.	(B)	188.	(A)	189.	(A)	190.	(C)	191.	(B)	192.	(D)
193.	(C)	194.	(B)	195.	(A)	196.	(A)	197.	(A)	198.	(A)
199.	(B)	200.	(D)	201.	(B)	202.	(C)	203.	(D)	204.	(D)
205.	(C)	206.	(D)	207.	(D)	208.	(D)	209.	(A)	210.	(C)
211.	(C)	212.	(A)	213.	(C)	214.	(D)	215.	(B)	216.	(B)
217.	(B)	218.	(B)	219.	(C)	220.	(C)	221.	(C)	222.	(A)
223.	(C)	224.	(ACD)	225.	(D)	226.	(A)	227.	(C)	228.	(C)
229.	()	230.	()	231.	(A)	232.	(C)	233.	(C)	234.	(B)
235.	(A)	236.	(C)	237.	(B)	238.	(A)	239.	(C)	240.	(D)
241.	(A)	242.	(A)	243.	(D)	244.	(B)	245.	(D)	246.	(B)
247.	(C)	248.	(C)	249.	(A)	250.	(C)	251.	(C)	252.	(A)
253.	(D)	254.	(A)	255.	(C)	256.	(A)	257.	(B)	258.	(A)
259.	(A)	260.	(C)	261.	(D)	262.	(C)	263.	(C)	264.	(B,C)
265.	(A)	266.	(B)	267.	(C)	268.	(A)	269.	(A)	270.	(A)
271.	(B)	272.	(C)	273.	(A)	274.	(C)	275.	(B)	276.	(A)
277.	(D)	278.	(A)	279.	(C)	280.	(B)	281.	(D)	282.	(A)
283.	(B)	284.	(B)	285.	(B)	286.	(C)	287.	(B)	288.	(A)
289.	(A)	290.	(B)	291.	(A)	292.	(BD)	293.	(C)	294.	(AB)
295.	(A)	296.	(B)	297.	(C)	298.	(B)	299.	(C)	300.	(B)